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Hierarchical PIwFF–Koopman-Based MPC for Offset-Free 6-DOF AUV Trajectory Tracking: Simulation and Pool Experiments

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ABSTRACT Accurate six-degree-of-freedom (6-DOF) trajectory tracking of autonomous underwater vehicles (AUVs) is challenging because nonlinear hydrodynamics, actuator constraints, and persistent plant–model mismatch degrade controller performance. This paper proposes a hierarchical PIwFF–Koopman MPC for offset-free 6-DOF trajectory tracking of the Xplorer-mini AUV. The outer PIwFF loop converts pose-tracking errors into body-fixed velocity commands, while the inner loop uses a Koopman-based linear prediction model identified with finite Koopman approximation and pool-specific biases, the inner MPC is augmented with an output-error integrator. DMDc and eDMDc are compared as prediction models; although eDMDc improves open-loop prediction, DMDc is selected for closed-loop control because it provides comparable tracking performance at a lower computational cost. A preview command is further used for time-varying references. Simulation and pool experiments on a figure-8 trajectory show that the proposed controller reduces steady-state bias, improves trajectory tracking relative to nominal Koopman MPC and PID–PID baselines.

INDEX TERMS Autonomous underwater vehicle (AUV), Koopman operator, model predictive control (MPC), data-driven control, trajectory tracking.

I. INTRODUCTION

Marine robotic vehicles support missions in oceans, gulfs, and inland waters—environmental monitoring, resource exploration, infrastructure inspection, and search-and-rescue—that are too vast or hazardous for crewed platforms. These missions are served by unmanned surface vehicles (USVs) such as autonomous sailboats [1], tethered remotely operated vehicles (ROVs) [2], and autonomous underwater vehicles (AUVs), with AUVs uniquely reaching subsea depths and ranges that surface craft cannot [3]. Across these missions, accurate trajectory tracking is fundamental to reliable navigation [4]—yet achieving it in practice remains an open problem rooted in the vehicle's own dynamics.

This difficulty stems from three coupled sources. First, the underlying six-degree-of-freedom (6-DOF) dynamics are highly nonlinear and cross-coupled: Coriolis and centripetal terms couple translation and rotation, quadratic hydrodynamic damping is speed-dependent, and sway motion induces yaw moments [5]. Second, effects such as buoyancy drift and sensor bias act as constant external disturbances over the control horizon [6]. Third, the inertial and hydrodynamic parameters fixed at design time drift through long-term wear and structural degradation, such as frame cracking from repeated operation, leaving a residual mismatch between the nominal model and the as-operated vehicle.

The Koopman operator transforms nonlinear dynamics into a linear evolution in an infinite-dimensional space of observables. When identified directly from data collected on the as-operated vehicle, the lifted model represents the actual rather than the nominal dynamics, thereby absorbing the parameter-mismatch source identified above at the identification stage. Two persistent biases nevertheless remain: a residual that arises because the operator must be approximated by a finite-dimensional model in practice [7], and the constant external disturbances identified above, which the identified model does not represent. Within the underwater domain, however, Koopman-based control has been applied only partially: for fault-tolerant backstepping on 8-thruster UVs [8], fractional sliding-mode tracking on 3-DOF AUVs [9], and delay-compensated observer control on AUVs [10]. None of these formulations rejects the persistent bias arising jointly from the Koopman approximation residual and constant external disturbance through an offset-free MPC formulation [11, 12], nor embeds such an offset-free Koopman MPC as the inner dynamic loop of a hierarchical trajectory-tracking architecture. To the best of our knowledge, no prior work has combined a kinematic outer loop with an offset-free Koopman-based linear MPC inner loop for 6-DOF AUV trajectory tracking, validated in real hardware.

To close this gap, we develop and validate a hierarchical PIwFF–Koopman MPC for 6-DOF AUV trajectory tracking on the Xplorer-mini AUV. The main contributions, presented in the order they appear in the paper, are:

- **Closed-loop system identification with DMDc and eDMDc.**
  An identification procedure using a diagonal-transit and sinusoidal-weaving maneuver designed to excite the cross-coupled 6-DOF dynamics, together with an empirical comparison of DMDc and eDMDc in both open-loop prediction and closed-loop tracking that reveals eDMDc to be more accurate in open loop yet DMDc to track better in closed loop, attributed to over-parameterization of the lifted basis.
- **Offset-free augmentation of the inner Koopman MPC with reference preview.**
  The inner linear MPC is augmented with an integral of the output tracking error in the lifted Koopman state, driving to zero the persistent bias arising jointly from the Koopman approximation residual and constant external disturbance, without requiring a separate observer. A preview component further anticipates time-varying references at the high-curvature points of a figure-8 trajectory.
- **Pool-experimental validation.**
  A comparative study of the proposed offset-free hierarchical controller against a nominal (non-offset-free) variant and a cascaded PID baseline in both simulation and pool experiments on the Xplorer-mini AUV.

The remainder of this paper is organized as follows. Section II reviews related work on AUV trajectory tracking, Koopman-based control, and offset-free MPC. Section III presents the preliminaries on the Koopman operator, DMDc, eDMDc, and MPC. Section IV describes the closed-loop system-identification protocol on the Xplorer-mini AUV and compares the identified DMDc and eDMDc models. Section V formulates the hierarchical PIwFF–Koopman MPC, comprising an outer kinematic PIwFF loop and an inner Koopman-based MPC in its nominal, offset-free, and preview-augmented forms. Section VI reports the simulation and pool experimental results. Section VII concludes the paper and outlines future directions.

II. RELATED WORK

AUV trajectory tracking has been addressed through several control families [3, 4, 6]. PID and cascaded schemes are practical but model-free and lack constraint handling. Sliding-mode control ensures robustness at the cost of chattering [13, 14]. Backstepping provides Lyapunov-based stability guarantees but depends on accurate hydrodynamic models. Adaptive control compensates parameter uncertainty online but complicates stability and tuning [15]. Nonlinear MPC handles constraints but is computationally heavy on embedded hardware [16]. A constraint-aware formulation independent of first-principles parameters thus remains an open need.

Unlike local Taylor linearization, the Koopman operator yields a *global* linear representation by lifting nonlinear dynamics into an infinite-dimensional observable space in which the evolution is exactly linear [17, 18]. Two complementary objects are typically sought: observables spanning a finite invariant subspace (directly used as a lifted linear model), and Koopman eigenfunctions (spectral modes that evolve as pure exponentials) [19, 20]. Two practical questions then arise: how to collect informative input–state data—passively or via active excitation [21, 22]—and how to choose a dictionary of observables. For autonomous systems, dynamic mode decomposition (DMD) uses the state itself as the dictionary [23], extended DMD (eDMD) adds hand-crafted bases (polynomials, RBFs) [24], kernel DMD employs a user-defined kernel [25], and deep autoencoders learn the embedding directly from data [26]. These methods extend to controlled systems through DMDc [27, 28], eDMDc [7, 29], and deep Koopman with inputs [30, 31], yielding linear surrogates directly compatible with linear MPC. Alternative routes such as sparse identification of nonlinear dynamics (SINDy) [32, 33] and physics-informed neural networks [34] remain inherently nonlinear and preclude direct linear control.

Because Koopman lifting yields linear dynamics, the full linear-control toolbox—pole placement, LQR, and most importantly for tracking, linear MPC with its convex formulation and explicit input/state constraints—applies directly to the original nonlinear system [7]. This combination has been demonstrated across a broad range of robotic platforms. On a pneumatic soft arm, Koopman MPC tracked references more than $3\times$ more accurately than an MPC built on a linearized model [35]. For quadrotors, an EDMDc model on $SE(3)$ enabled a 100 Hz linear MPC to track agile references [36]. For autonomous ground vehicles, a deep Koopman–MPC outperformed nonlinear and linear time-varying MPC baselines in CarSim simulations [37]. A derivative-based Koopman model with LQR controlled a tail-actuated robotic fish, outperforming a tuned PID [38]. Additional robotic applications are surveyed in [22].

Offset-free MPC achieves zero steady-state tracking error under plant–model mismatch by augmenting the prediction model with a constant disturbance state estimated online [11, 12]; more generally, the augmentation rejects any persistent bias at the output. For our Koopman MPC, parameter mismatch is already absorbed by data-driven identification on the as-operated vehicle, leaving two persistent biases on the lifted dynamics—the finite-dimensional approximation residual and constant external disturbances such as buoyancy drift and sensor bias—both absorbed by a single augmented disturbance model. This combination has not yet been validated as the inner dynamic loop of a hierarchical AUV trajectory-tracking controller in pool experiments, and is the focus of this work.

III. PRELIMINARIES

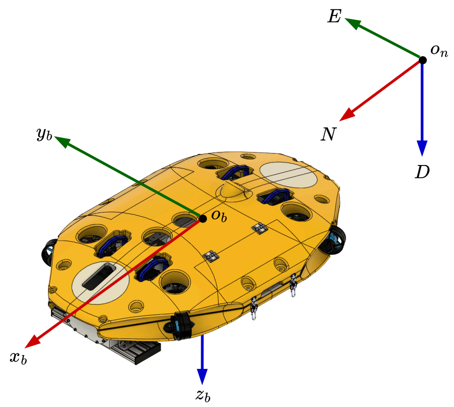

FIGURE 1. Xplorer-mini

A. AUV MODEL

This section presents the AUV dynamic model, which serves as the physical foundation for the Koopman-based linear model developed in this work. The vehicle motion is described in six degrees of freedom (6-DOF), including translational motions in surge, sway, and heave, and rotational motions in roll, pitch, and yaw, following the standard marine craft modeling framework in [5].

Let the pose of the AUV expressed in the NED frame be defined as

$$\boldsymbol{\eta} = [x,\ y,\ z,\ \phi,\ \theta,\ \psi]^\top \in \mathbb{R}^6, \tag{1}$$

and the body-fixed velocity vector as

$$\boldsymbol{\nu} = [u,\ v,\ w,\ p,\ q,\ r]^\top \in \mathbb{R}^6. \tag{2}$$

The kinematic model relates the body-fixed velocity to the rate of change of the vehicle pose in the inertial frame as

$$\dot{\boldsymbol{\eta}} = \boldsymbol{J}(\boldsymbol{\eta})\boldsymbol{\nu}, \tag{3}$$

where $\boldsymbol{J}(\boldsymbol{\eta}) \in \mathbb{R}^{6\times6}$ is the transformation, given by

$$\boldsymbol{J}(\boldsymbol{\eta}) = \begin{bmatrix} \boldsymbol{R}(\boldsymbol{\eta}) & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{T}(\boldsymbol{\eta}) \end{bmatrix}. \tag{4}$$

Here, $\boldsymbol{R}(\boldsymbol{\eta}) \in \mathbb{R}^{3\times3}$ is the rotation matrix from the body-fixed frame to the inertial frame, and $\boldsymbol{T}(\boldsymbol{\eta}) \in \mathbb{R}^{3\times3}$ maps the angular velocity vector to the Euler-angle rates. The explicit forms of $\boldsymbol{R}(\boldsymbol{\eta})$ and $\boldsymbol{T}(\boldsymbol{\eta})$ follow the standard 6-DOF marine craft model in [5].

For control-oriented modeling, the current-induced hydrodynamic effects and modeling uncertainties are grouped into an equivalent generalized disturbance torque $\boldsymbol{\tau}_d \in \mathbb{R}^6$. Therefore, the AUV dynamics are rewritten as

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} + \boldsymbol{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} + \boldsymbol{\tau}_d. \tag{5}$$

where $\boldsymbol{M} \in \mathbb{R}^{6\times6}$ is the inertia matrix, $\boldsymbol{C}(\boldsymbol{\nu}_r) \in \mathbb{R}^{6\times6}$ is the Coriolis and centripetal matrix, $\boldsymbol{D}(\boldsymbol{\nu}_r) \in \mathbb{R}^{6\times6}$ is the hydrodynamic damping matrix, $\boldsymbol{g}(\boldsymbol{\eta})$ denotes hydrostatic restoring terms, and $\boldsymbol{\tau} \in \mathbb{R}^6$ is the generalized control torque.

The nominal dynamic model is then defined by neglecting the disturbance term $\boldsymbol{\tau}_d$, yielding

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} + \boldsymbol{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{g}(\boldsymbol{\eta}) = \boldsymbol{\tau}. \tag{6}$$

In this work, the nominal model in (6) provides the physical insight that guides the construction of the Koopman-based linear model and serves as the plant for the numerical simulation. The explicit expressions of $\boldsymbol{M}$, $\boldsymbol{C}(\boldsymbol{\nu})$, $\boldsymbol{D}(\boldsymbol{\nu})$, $\boldsymbol{g}(\boldsymbol{\eta})$, and the corresponding parameter values for the Xplorer-mini AUV are given in Appendix A. The effect of ocean currents is treated as an external disturbance.

B. KOOPMAN-BASED LIFTED LINEAR MODEL

In this work, the Koopman-based lifted linear model is constructed using the nominal velocity dynamics, obtained from (6), as

$$\begin{aligned} \dot{\boldsymbol{\nu}} &= \boldsymbol{f}_\nu(\boldsymbol{\eta}, \boldsymbol{\nu}, \boldsymbol{\tau}) \\ &= \boldsymbol{M}^{-1}(\boldsymbol{\tau} - \boldsymbol{C}(\boldsymbol{\nu})\boldsymbol{\nu} - \boldsymbol{D}(\boldsymbol{\nu})\boldsymbol{\nu} - \boldsymbol{g}(\boldsymbol{\eta})). \end{aligned} \tag{7}$$

By discretizing (7) with sampling period T_s, the corresponding nominal discrete-time velocity dynamics are given by

$$\boldsymbol{\nu}_{k+1} = \boldsymbol{F}(\boldsymbol{\nu}_k, \boldsymbol{\tau}_k), \tag{8}$$

where $\boldsymbol{\nu}_k = \boldsymbol{\nu}(kT_s)$ and $\boldsymbol{\tau}_k = \boldsymbol{\tau}(kT_s)$.

Remark 1. Notice that the nominal velocity dynamics in (7) depends on the pose $\boldsymbol{\eta}$ through the hydrostatic restoring term $\boldsymbol{g}(\boldsymbol{\eta})$. In this work, because the AUV operates under closed-loop control, it maintains a tightly regulated pose. Consequently, variations in $\boldsymbol{g}(\boldsymbol{\eta})$ are treated as bounded unmodeled dynamics, allowing the pose dependency to be omitted from the nominal discrete-time mapping in (8). These variations are later absorbed into the lumped disturbance vector. □

Let the standardized velocity vector $\bar{\boldsymbol{\nu}}_k$ and standardized control torque vector $\bar{\boldsymbol{\tau}}_k$ be defined as

$$\bar{\boldsymbol{\nu}}_k = \boldsymbol{\Sigma}_\nu^{-1}(\boldsymbol{\nu}_k - \boldsymbol{\mu}_\nu), \qquad \bar{\boldsymbol{\tau}}_k = \boldsymbol{\Sigma}_\tau^{-1}(\boldsymbol{\tau}_k - \boldsymbol{\mu}_\tau), \tag{9}$$

where $\boldsymbol{\mu}_\nu$ and $\boldsymbol{\mu}_\tau$ are the sample mean vectors, and $\boldsymbol{\Sigma}_\nu$ and $\boldsymbol{\Sigma}_\tau$ are the diagonal sample standard deviation matrices associated with the velocity and control torque data, respectively.

To construct the Koopman-based lifted linear model, let

$$\psi_i : \bar{\mathcal{V}} \to \mathbb{R}, \qquad i = 1, 2, \ldots, N_z,$$

denote observable basis functions, where $\bar{\mathcal{V}} \subseteq \mathbb{R}^6$ represents the domain of the standardized velocities, and N_z is the number of dictionary elements. These basis functions define the lifting map

$$\boldsymbol{z} = \boldsymbol{\psi}(\bar{\boldsymbol{\nu}}) = \begin{bmatrix} \psi_1(\bar{\boldsymbol{\nu}}) \\ \psi_2(\bar{\boldsymbol{\nu}}) \\ \vdots \\ \psi_{N_z}(\bar{\boldsymbol{\nu}}) \end{bmatrix} \in \mathbb{R}^{N_z}, \tag{10}$$

where $\boldsymbol{z}$ denotes the lifted state vector. The choice of the dictionary determines the expressive power of the lifted model and may include the original states together with polynomial, radial basis, or physics-informed nonlinear terms.

Using a finite-dimensional approximation of the Koopman operator, the nonlinear velocity dynamics in (8) are represented in the lifted space by the nominal linear model as

$$\boldsymbol{z}_{k+1} = \boldsymbol{A}\boldsymbol{z}_k + \boldsymbol{B}\bar{\boldsymbol{\tau}}_k, \tag{11}$$

where $\boldsymbol{A} \in \mathbb{R}^{N_z \times N_z}$ and $\boldsymbol{B} \in \mathbb{R}^{N_z \times 6}$ are the lifted system and input matrices, respectively.

The standardized velocity vector can be recovered from the lifted state vector by

$$\bar{\boldsymbol{\nu}}_k = \boldsymbol{C}\boldsymbol{z}_k, \tag{12}$$

where $\boldsymbol{C} \in \mathbb{R}^{6 \times N_z}$ is the output matrix.

Remark 2. Since (11) approximates the nominal nonlinear dynamics, there is always some inherent approximation error. Moreover, the AUV is also affected by ocean-current-induced loads, unmodeled hydrodynamic forces, and other external disturbances. To account for these effects, the lifted linear model can be extended to a disturbance-augmented form as

$$\boldsymbol{z}_{k+1} = \boldsymbol{A}\boldsymbol{z}_k + \boldsymbol{B}\bar{\boldsymbol{\tau}}_k + \boldsymbol{w}_k, \tag{13}$$

where $\boldsymbol{w}_k \in \mathbb{R}^{N_z}$ denotes the lumped disturbance vector, representing approximation errors, ocean-current effects, residual hydrodynamic uncertainty, and other environmental disturbances. □

Assumption 1. There exists $w_{\max} \in \mathbb{R}$ such that

$$\|\boldsymbol{w}_k\| \leq w_{\max}, \tag{14}$$

i.e., $\boldsymbol{w}_k$ is bounded. □

IV. SYSTEM IDENTIFICATION VIA DMDC AND EDMDC METHODS

A. DATA COLLECTION AND SNAPSHOT MATRICES

The concept of closed-loop system identification is shown in Fig. 2. Accurate identification requires a dataset that sufficiently excites the AUV's nonlinear, cross-coupled dynamics. To achieve this, a sinusoidal weaving maneuver is adopted to continuously excite the sway-yaw dynamics, thereby revealing the coupling terms in the hydrodynamic and Coriolis matrices.

Let the initial and terminal pose vectors be defined as

$$\boldsymbol{\eta}_0 = [x_0, y_0, z_0, \phi_0, \theta_0, \psi_0]^\top, \tag{15}$$

$$\boldsymbol{\eta}_f = [x_f, y_f, z_f, \phi_f, \theta_f, \psi_f]^\top, \tag{16}$$

respectively. The corresponding pose displacement vector is

$$\boldsymbol{d}_{\boldsymbol{\eta}} = \boldsymbol{\eta}_f - \boldsymbol{\eta}_0 = [d_x, d_y, d_z, d_\phi, d_\theta, d_\psi]^\top.$$

Based on the horizontal displacement components, the unit vector perpendicular to the translation direction is defined as

$$\hat{v}_x = -\frac{d_y}{\sqrt{d_x^2 + d_y^2}}, \quad \hat{v}_y = \frac{d_x}{\sqrt{d_x^2 + d_y^2}},$$

and the weaving angular frequency is given by

$$\omega = \frac{2\pi n_{zig}}{t_f}, \tag{17}$$

where n_{zig} denotes the number of weaving cycles over the maneuver duration t_f. The corresponding reference velocity components on the horizontal plane are

$$\dot{x}_d(t) = \frac{d_x}{t_f} + A\omega\cos(\omega t)\hat{v}_x$$

$$\dot{y}_d(t) = \frac{d_y}{t_f} + A\omega\cos(\omega t)\hat{v}_y$$

where A is the weaving amplitude. The resulting time-varying reference pose vector $\boldsymbol{\eta}_{\mathrm{ref}}(t) \in \mathbb{R}^6$ is then defined as

$$\boldsymbol{\eta}_{\mathrm{ref}}(t) = \begin{bmatrix} x_0 + \frac{t}{t_f}d_x + \hat{v}_x A\sin(\omega t) \\ y_0 + \frac{t}{t_f}d_y + \hat{v}_y A\sin(\omega t) \\ z_0 + \frac{t}{t_f}d_z \\ \phi_0 + \frac{t}{t_f}d_\phi \\ \theta_0 + \frac{t}{t_f}d_\theta \\ \operatorname{atan2}\left(\dot{y}_d(t), \dot{x}_d(t)\right) \end{bmatrix}. \tag{18}$$

To enrich the dataset for closed-loop system identification, the vehicle was controlled using a cascaded PID-PID architecture, and the generalized control input was perturbed by zero-mean Gaussian white noise,

$$\boldsymbol{\tau}_{\mathrm{prt}} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2\boldsymbol{I}) \in \mathbb{R}^6. \tag{19}$$

The commanded generalized control input is therefore given by

$$\boldsymbol{\tau}_{\mathrm{cmd}} = \boldsymbol{\tau}_{\mathrm{pid}} + \boldsymbol{\tau}_{\mathrm{prt}}. \tag{20}$$

Assume that $N_s + 1$ samples of the body-fixed velocity and generalized control torque are collected and standardized, yielding N_s one-step snapshot triplets

$$\{\bar{\boldsymbol{\nu}}_k,\ \bar{\boldsymbol{\tau}}_k,\ \bar{\boldsymbol{\nu}}_{k+1}\}_{k=0}^{N_s-1}. \tag{21}$$

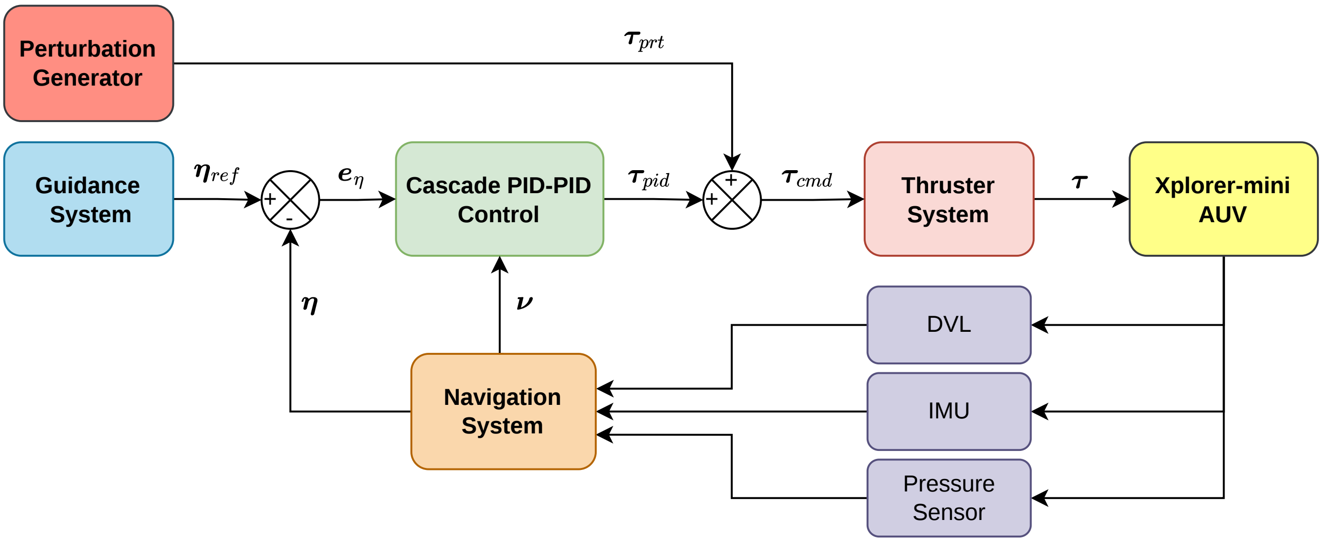

FIGURE 2. **Closed-loop system identification framework for the Xplorer-mini AUV using cascade PID–PID control, state feedback from the navigation system, and perturbation injection for excitation enrichment.**

The corresponding snapshot matrices are constructed as

$$\boldsymbol{X} = \begin{bmatrix} | & | & & | \\ \bar{\boldsymbol{\nu}}_0 & \bar{\boldsymbol{\nu}}_1 & \cdots & \bar{\boldsymbol{\nu}}_{N_s-1} \\ | & | & & | \end{bmatrix}, \tag{22}$$

$$\boldsymbol{X}^+ = \begin{bmatrix} | & | & & | \\ \bar{\boldsymbol{\nu}}_1 & \bar{\boldsymbol{\nu}}_2 & \cdots & \bar{\boldsymbol{\nu}}_{N_s} \\ | & | & & | \end{bmatrix}, \tag{23}$$

$$\boldsymbol{U} = \begin{bmatrix} | & | & & | \\ \bar{\boldsymbol{\tau}}_0 & \bar{\boldsymbol{\tau}}_1 & \cdots & \bar{\boldsymbol{\tau}}_{N_s-1} \\ | & | & & | \end{bmatrix}. \tag{24}$$

B. UNIFIED KOOPMAN-BASED LINEAR MODEL

To construct a data-driven finite-dimensional approximation of the Koopman-based linear model, this study employs two methods: dynamic mode decomposition with control (DMDc) and extended dynamic mode decomposition with control (eDMDc).

Both methods are formulated within the unified Koopman-based linear model framework defined in (11), where DMDc is treated as a special case of eDMDc utilizing the identity lifting map, i.e., $\psi(\bar{\boldsymbol{\nu}}) = \bar{\boldsymbol{\nu}}$.

1) Koopman-Based Linear Model via DMDc

For DMDc, substituting $z_k = \bar{\boldsymbol{\nu}}_k$ into (11) yields

$$\bar{\boldsymbol{\nu}}_{k+1} \approx \boldsymbol{A}\bar{\boldsymbol{\nu}}_k + \boldsymbol{B}\bar{\boldsymbol{\tau}}_k. \tag{25}$$

The nominal DMDc regression problem is

$$\boldsymbol{X}^+ \approx \boldsymbol{A}\boldsymbol{X} + \boldsymbol{B}\boldsymbol{U}. \tag{26}$$

Then, the DMDc model matrices are obtained from the least-squares problem

$$\min_{\boldsymbol{A},\boldsymbol{B}} \left\| \boldsymbol{X}^+ - \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} \boldsymbol{X} \\ \boldsymbol{U} \end{bmatrix} \right\|_F^2, \tag{27}$$

with solution

$$\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \end{bmatrix} = \boldsymbol{X}^+ \begin{bmatrix} \boldsymbol{X} \\ \boldsymbol{U} \end{bmatrix}^\dagger, \tag{28}$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse.

2) Koopman-Based Lifted Linear Model via eDMDc

For eDMDc, the state is defined in the lifted observable space using the lifting map introduced in (10). Specifically,

$$z_k = \psi(\bar{\boldsymbol{\nu}}_k) \in \mathbb{R}^{N_z}. \tag{29}$$

The lifted dynamics are then approximated by the same unified linear model form

$$z_{k+1} \approx \boldsymbol{A}z_k + \boldsymbol{B}\bar{\boldsymbol{\tau}}_k. \tag{30}$$

To identify the lifted model, the state snapshots are mapped into the observable space:

$$\boldsymbol{Z} = \begin{bmatrix} | & | & & | \\ z_0 & z_1 & \cdots & z_{N_s-1} \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \psi(\bar{\boldsymbol{\nu}}_0) & \psi(\bar{\boldsymbol{\nu}}_1) & \cdots & \psi(\bar{\boldsymbol{\nu}}_{N_s-1}) \\ | & | & & | \end{bmatrix}, \tag{31}$$

$$\boldsymbol{Z}^+ = \begin{bmatrix} | & | & & | \\ z_1 & z_2 & \cdots & z_{N_s} \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \psi(\bar{\boldsymbol{\nu}}_1) & \psi(\bar{\boldsymbol{\nu}}_2) & \cdots & \psi(\bar{\boldsymbol{\nu}}_{N_s}) \\ | & | & & | \end{bmatrix}. \tag{32}$$

The eDMDc regression problem is formulated as

$$\boldsymbol{Z}^+ \approx \boldsymbol{A}\boldsymbol{Z} + \boldsymbol{B}\boldsymbol{U}. \tag{33}$$

The lifted model matrices are obtained from

$$\min_{\boldsymbol{A},\boldsymbol{B}} \left\| \boldsymbol{Z}^{+} - [\boldsymbol{A} \quad \boldsymbol{B}] \begin{bmatrix} \boldsymbol{Z} \\ \boldsymbol{U} \end{bmatrix} \right\|_F^2, \tag{34}$$

with solution

$$[\boldsymbol{A} \quad \boldsymbol{B}] = \boldsymbol{Z}^{+} \begin{bmatrix} \boldsymbol{Z} \\ \boldsymbol{U} \end{bmatrix}^{\dagger}. \tag{35}$$

Remark 3. In this work, a second-order polynomial basis is selected to approximate the nonlinear hydrodynamic damping and Coriolis effects:

$$z = \psi(\bar{\boldsymbol{\nu}}) = [\bar{\nu}_i,\ \bar{\nu}_i\bar{\nu}_j]^{\top} \in \mathbb{R}^{27}, \quad 1 \le i \le j \le 6.$$

This 27-dimensional lifted state enables the eDMDc framework to linearly capture the inherent quadratic drag and cross-coupled motions. □

Assumption 2. The pair $(\boldsymbol{A}, \boldsymbol{B})$ derived from the data-driven Koopman operator identification are stabilizable. □

In this work, the pair $(\boldsymbol{A}, \boldsymbol{B})$'s of Koopman-based linear models via both DMDc and eDMDc are stabilizable.

V. PIWFF–KOOPMAN-BASED MPC FRAMEWORK

Based on the linear models identified using DMDc and eDMDc, a PIwFF–Koopman-based MPC framework shown in Fig. 3 is proposed for AUV pose tracking. A cascade control architecture is adopted, in which an outer-loop PIwFF controller generates the body-fixed velocity command from the pose-tracking error. At the same time, an inner-loop Koopman-based MPC determines the control torques subject to actuator and state constraints.

A. PIWFF CONTROLLER

Let the bounded reference pose trajectory at sampling instant k be denoted by

$$\boldsymbol{\eta}_{\mathrm{ref},k} = [x_{\mathrm{ref},k},\ y_{\mathrm{ref},k},\ z_{\mathrm{ref},k},\ \phi_{\mathrm{ref},k},\ \theta_{\mathrm{ref},k},\ \psi_{\mathrm{ref},k}]^{\top}. \tag{36}$$

The pose-tracking error and its integral state are defined as

$$\boldsymbol{e}_{\boldsymbol{\eta},k} = \boldsymbol{\eta}_{\mathrm{ref},k} - \boldsymbol{\eta}_k, \tag{37}$$

$$\boldsymbol{\xi}_{\boldsymbol{\eta},k} = \boldsymbol{\xi}_{\boldsymbol{\eta},k-1} + T_s \boldsymbol{e}_{\boldsymbol{\eta},k-1} \tag{38}$$

with $\boldsymbol{\xi}_{\boldsymbol{\eta},0} = \boldsymbol{0}$.

Remark 4. To prevent integral windup during sustained trajectory deviations or actuator saturation, the integral update law is implemented with the following componentwise clamping mechanism:

$$\boldsymbol{\xi}_{\boldsymbol{\eta},k} = \mathrm{sat}_{[\boldsymbol{\xi}_{\boldsymbol{\eta},\min},\boldsymbol{\xi}_{\boldsymbol{\eta},\max}]} \left(\boldsymbol{\xi}_{\boldsymbol{\eta},k-1} + T_s \boldsymbol{e}_{\boldsymbol{\eta},k-1} \right),$$

where $\boldsymbol{\xi}_{\boldsymbol{\eta},\min}$ and $\boldsymbol{\xi}_{\boldsymbol{\eta},\max}$ are the minimum and maximum of the integrator state. □

The outer-loop controller generates the velocity command in the body frame from the pose-tracking error and the reference pose rate. This work utilizes a PIwFF structure:

$$\boldsymbol{\nu}_{\mathrm{cmd},k} = \boldsymbol{J}^{-1}(\boldsymbol{\eta}_k) \underbrace{\left(\dot{\boldsymbol{\eta}}_{\mathrm{ref,k}} + \boldsymbol{K}_{P,\boldsymbol{\eta}}\, \boldsymbol{e}_{\boldsymbol{\eta},k} + \boldsymbol{K}_{I,\boldsymbol{\eta}}\, \boldsymbol{\xi}_{\boldsymbol{\eta},k} \right)}_{\dot{\boldsymbol{\eta}}_{\mathrm{cmd},k}}. \tag{39}$$

where $\boldsymbol{K}_{P,\boldsymbol{\eta}} \succ 0$ and $\boldsymbol{K}_{I,\boldsymbol{\eta}} \succ 0$ are positive definite gain matrices.

This instantaneous velocity command serves as the reference for the inner-loop Koopman-based MPC.

Remark 5. In this work, Xplorer-mini AUV operates under closed-loop control, ensuring that the pitch angle remains bounded within $\theta \in [-\theta_{\mathrm{oper}}, \theta_{\mathrm{oper}}]$ for some operational limit $\theta_{\mathrm{oper}} < 90^{\circ}$. Thus, the inverse Jacobian matrix $\boldsymbol{J}^{-1}$ is always well-defined. □

B. LIFTED-STATE COMMAND GENERATION

Since the inner-loop MPC requires a sequence of velocity and lifted-state commands over the prediction horizon, the outer-loop PIwFF controller is evaluated recursively to generate this sequence. There are two methods: non-preview and preview command generation.

1) Non-Preview Command Generation

For non-preview command generation, the command is kept constant over the predicted horizon N_p:

$$\boldsymbol{\nu}_{\mathrm{cmd},k+i|k} = \boldsymbol{\nu}_{\mathrm{cmd},k}, \tag{40}$$

for all $i = 0, 1, \ldots, N_p - 1$.

2) Preview Command Generation

To generate preview commands, the initial conditions at time step k are defined as:

$$\boldsymbol{\nu}_{\mathrm{cmd},k|k} = \boldsymbol{\nu}_{\mathrm{cmd},k}, \tag{41}$$

$$\boldsymbol{\eta}_{k|k} = \boldsymbol{\eta}_k, \tag{42}$$

$$\boldsymbol{\xi}_{\boldsymbol{\eta},k|k} = \boldsymbol{\xi}_{\boldsymbol{\eta},k}. \tag{43}$$

Within the recursive prediction horizon, the i-step-ahead predicted pose is first obtained using a forward Euler approximation:

$$\boldsymbol{\eta}_{k+i|k} = \boldsymbol{\eta}_{k+i-1|k} + T_s \boldsymbol{J}\left(\boldsymbol{\eta}_{k+i-1|k}\right) \boldsymbol{\nu}_{\mathrm{cmd},k+i-1|k}. \tag{44}$$

Remark 6. Note that the forward Euler approximation in (44) propagates the kinematics using the commanded velocity rather than the predicted actual velocity. This formulation inherently assumes that the inner-loop Koopman MPC achieves sufficiently fast and accurate tracking relative to the outer-loop kinematic bandwidth. □

The predicted pose-tracking error and its integral state are computed as

$$\boldsymbol{e}_{\boldsymbol{\eta},k+i|k} = \boldsymbol{\eta}_{\mathrm{ref},k+i} - \boldsymbol{\eta}_{k+i|k}, \tag{45}$$

$$\boldsymbol{\xi}_{\boldsymbol{\eta},k+i|k} = \boldsymbol{\xi}_{\boldsymbol{\eta},k+i-1|k} + T_s \boldsymbol{e}_{\boldsymbol{\eta},k+i-1|k}. \tag{46}$$

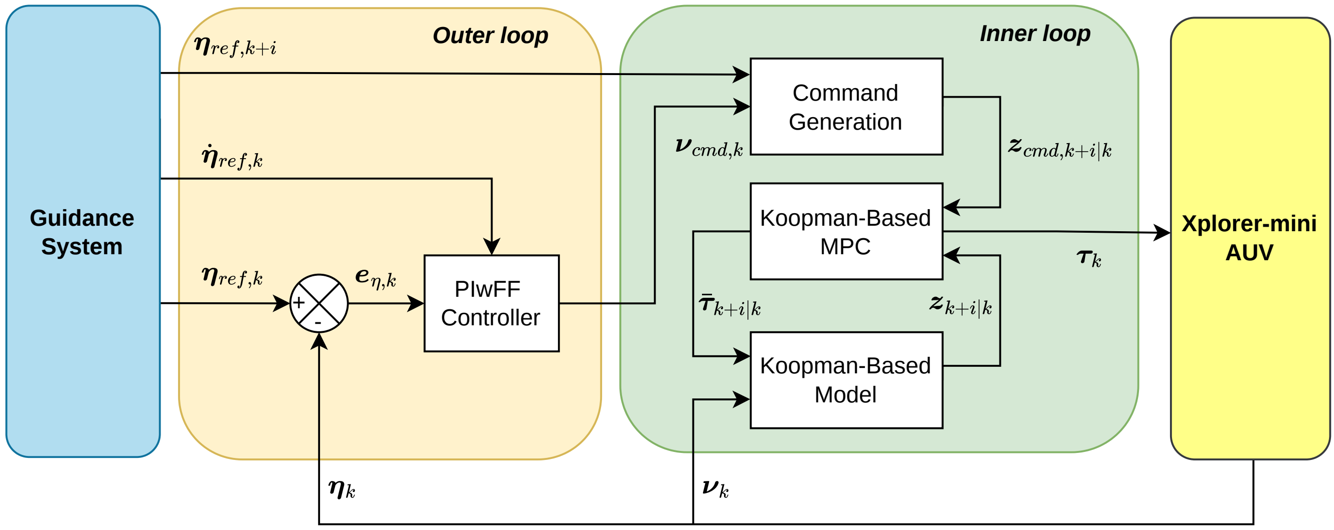

FIGURE 3. PIwFF–Koopman-based MPC block diagram. Cascade control architecture of the proposed PIwFF–Koopman-based MPC framework for the Xplorer-mini AUV, consisting of an outer-loop PIwFF kinematic controller, command generation block, inner-loop Koopman-based MPC, and Koopman-based prediction model.

Remark 7. Practically, the integral windup prevention by the clamping mechanism is used:

$$\xi_{\eta,k+i|k} = \mathrm{sat}_{[\xi_{\eta,\min},\xi_{\eta,\max}]}\left(\xi_{\eta,k+i-1|k} + T_s e_{\eta,k+i-1|k}\right).$$

Finally, the i-step-ahead preview velocity command is computed via the inverse kinematics:

$$\nu_{\mathrm{cmd},k+i|k} = J^{-1}(\eta_{k+i|k})\Big(\dot{\eta}_{\mathrm{ref},k+i} + K_{P,\eta} e_{\eta,k+i|k} + K_{I,\eta}\xi_{\eta,k+i|k}\Big). \tag{47}$$

3) Unified Lifted-State Command Generation

To ensure that both non-preview and preview velocity commands remain within admissible bounds, they are calculated using the saturation function as

$$\nu^{\mathrm{sat}}_{\mathrm{cmd},k+i|k} = \mathrm{sat}_{[\nu_{\min},\nu_{\max}]}\left(\nu_{\mathrm{cmd},k+i|k}\right), \tag{48}$$

where $\nu_{\min}$ and $\nu_{\max}$ denote the lower and upper velocity limits, respectively.

Remark 8. In this work, the overall admissible velocity bounds of Xplorer-mini AUV were defined as

$$|u|, |v|, |w| \leq 1 \text{ m/s}, \qquad |p|, |q|, |r| \leq 1 \text{ rad/s},$$

which represent broad modeling bounds rather than task-executed operating limits. □

The velocity commands are then standardized using the predefined statistics:

$$\bar{\nu}_{\mathrm{cmd},k+i|k} = \Sigma_\nu^{-1}\left(\nu^{\mathrm{sat}}_{\mathrm{cmd},k+i|k} - \mu_\nu\right). \tag{49}$$

Finally, the Koopman lifting function maps the standardized command into the higher-dimensional observable space to obtain the lifted-state commands as

$$z_{\mathrm{cmd},k+i|k} = \psi(\bar{\nu}_{\mathrm{cmd},k+i|k}). \tag{50}$$

These preconditioning steps are necessary before lifting to avoid unbounded basis functions and to ensure stable real-time computation.

C. KOOPMAN-BASED MPC DESIGN

The Koopman-based MPC is designed to track the lifted-state commands while penalizing the standardized control-torque effort, subject to velocity and control-torque constraints, over a prediction horizon of N_p. The constraint sets are defined as

$$\mathbb{Z} = \left\{z \in \mathbb{R}^{N_z} \,\middle|\, \bar{\nu}_{\min} \leq Cz \leq \bar{\nu}_{\max}\right\}, \tag{51}$$

$$\bar{\mathbb{T}} = \left\{\bar{\tau} \in \mathbb{R}^6 \,\middle|\, \bar{\tau}_{\min} \leq \bar{\tau} \leq \bar{\tau}_{\max}\right\}. \tag{52}$$

1) Nominal Koopman-based MPC

To achieve precise trajectory tracking, the standard MPC is formulated as a finite-horizon optimal control problem at each sampling instant k:

$$\begin{aligned} \underset{\{\bar{\tau}_{k+i|k}\}_{i=0}^{N_p-1}}{\text{minimize}} \quad & J = \sum_{i=0}^{N_p-1} \ell(e_{z,k+i|k}, \bar{\tau}_{k+i|k}) + V_f(e_{z,k+N_p|k}) \\ \text{subject to} \quad & z_{k+i+1|k} = A z_{k+i|k} + B\bar{\tau}_{k+i|k}, \\ & z_{k|k} = z_k, \\ & z_{k+i|k} \in \mathbb{Z}, \quad \bar{\tau}_{k+i|k} \in \bar{\mathbb{T}}, \\ & z_{k+N_p|k} \in \mathbb{Z}_f \subseteq \mathbb{Z} \end{aligned} \tag{53}$$

where $e_{z,k+i|k} = z_{k+i|k} - z_{\mathrm{cmd},k+i|k}$ denotes the tracking error. The quadratic stage and terminal costs are defined as $\ell(e_z, \bar{\tau}) = e_z^\top Q e_z + \bar{\tau}^\top R \bar{\tau}$ and $V_f(e_z) = e_z^\top P e_z$, respectively, with weighting matrices $Q \succeq 0$ and $R \succ 0$.

The existence of a unique terminal penalty matrix $P \succ 0$ is guaranteed by solving the Discrete Algebraic Riccati Equation (DARE):

$$\boldsymbol{A}^\top\boldsymbol{P}\boldsymbol{A}-\boldsymbol{P}-\boldsymbol{A}^\top\boldsymbol{P}\boldsymbol{B}(\boldsymbol{R}+\boldsymbol{B}^\top\boldsymbol{P}\boldsymbol{B})^{-1}\boldsymbol{B}^\top\boldsymbol{P}\boldsymbol{A}+\boldsymbol{Q}=\boldsymbol{0}. \quad (54)$$

The terminal set $\mathbb{Z}_f$ is defined as the MPI set under the optimal LQR law $\bar{\boldsymbol{\tau}}=-\boldsymbol{K}\boldsymbol{e}_z$, where the gain is $\boldsymbol{K}=(\boldsymbol{R}+\boldsymbol{B}^\top\boldsymbol{P}\boldsymbol{B})^{-1}\boldsymbol{B}^\top\boldsymbol{P}\boldsymbol{A}$.

Assumption 3. The nominal Koopman-based MPC is feasible, and $\bar{\boldsymbol{\tau}}=-\boldsymbol{K}\boldsymbol{e}_z\in\bar{\mathbb{T}}$ to ensure recursive feasibility and local stability. □

Following the receding horizon principle, only the first element of the optimal sequence $\{\bar{\boldsymbol{\tau}}^*_{k+i|k}\}_{i=0}^{N_p-1}$ is applied to the system, i.e., $\bar{\boldsymbol{\tau}}_k=\bar{\boldsymbol{\tau}}^*_{k|k}$. Finally, $\bar{\boldsymbol{\tau}}_k$ is destandardized to obtain the generalized control torque as

$$\boldsymbol{\tau}_k=\boldsymbol{\Sigma}_\tau\,\bar{\boldsymbol{\tau}}_k+\boldsymbol{\mu}_\tau. \quad (55)$$

This generalized control torque is then sent to the AUV.

2) Offset-Free Koopman-based MPC

While the nominal MPC provides optimal performance, external disturbances and model mismatches may lead to steady-state tracking offsets. To ensure offset-free tracking, the prediction model is augmented with an integral error state $\boldsymbol{\xi}_z\in\mathbb{R}^6$, which evolves as

$$\boldsymbol{\xi}_{z,k+i+1|k}=\boldsymbol{\xi}_{z,k+i|k}+T_s\boldsymbol{C}\boldsymbol{e}_{z,k+i|k} \quad (56)$$

where T_s denotes the sampling time.

Remark 9. In this work, the integral windup prevention by the clamping mechanism is used:

$$\boldsymbol{\xi}_{z,k+i+1|k}=\mathrm{sat}_{[\boldsymbol{\xi}_{z,\min},\boldsymbol{\xi}_{z,\max}]}\left(\boldsymbol{\xi}_{z,k+i|k}+T_s\boldsymbol{C}\boldsymbol{e}_{z,k+i|k}\right).$$

where $\boldsymbol{\xi}_{z,\min}$ and $\boldsymbol{\xi}_{z,\max}$ are the minimum and maximum of the integrator error state. □

By defining the augmented state vector $\tilde{z}_{k+i|k}=[z_{k+i|k}^\top,\ \boldsymbol{\xi}_{z,k+i|k}^\top]^\top$, the augmented prediction model is formulated as

$$\tilde{z}_{k+i+1|k}=\underbrace{\begin{bmatrix}\boldsymbol{A}&\boldsymbol{0}\\T_s\boldsymbol{C}&\boldsymbol{I}\end{bmatrix}}_{\tilde{\boldsymbol{A}}}\tilde{z}_{k+i|k}+\underbrace{\begin{bmatrix}\boldsymbol{B}\\\boldsymbol{0}\end{bmatrix}}_{\tilde{\boldsymbol{B}}}\bar{\boldsymbol{\tau}}_{k+i|k}+\underbrace{\begin{bmatrix}\boldsymbol{0}\\-T_s\boldsymbol{C}\end{bmatrix}}_{\tilde{\boldsymbol{E}}}z_{\mathrm{cmd},k+i|k}, \quad (57)$$

where $\boldsymbol{I}$ and $\boldsymbol{0}$ represent the identity and zero matrices of appropriate dimensions, respectively.

Assumption 4. The pair $(\tilde{\boldsymbol{A}},\tilde{\boldsymbol{B}})$ to be stabilizable. □

Remark 10. According to the Popov-Belevitch-Hautus (PBH) test for integral control, this condition is satisfied provided that $(\boldsymbol{A},\boldsymbol{B})$ is stabilizable, and the system matrix has no invariant zeros at $z=1$. Mathematically, this dictates the rank condition:

$$\mathrm{rank}\begin{bmatrix}\boldsymbol{A}-\boldsymbol{I}&\boldsymbol{B}\\T_s\boldsymbol{C}&\boldsymbol{0}\end{bmatrix}=N_z+\dim(\boldsymbol{\xi}_z),$$

To fulfill this rank condition and prevent the loss of controllability in high-dimensional lifted systems (where $N_z>\dim(\boldsymbol{\tau})$), the integral state augmentation $\boldsymbol{\xi}_z$ must be judiciously restricted such that the number of integrators does not exceed the number of independent control inputs. □

The objective of the integral MPC (i.e., offset-free MPC) is to accurately track the lifted-state command while driving the accumulated integral error to zero. To this end, the offset-free MPC is defined as

$$\begin{aligned}\underset{\{\bar{\boldsymbol{\tau}}_{k+i|k}\}_{i=0}^{N_p-1}}{\text{minimize}}\quad & J=\sum_{i=0}^{N_p-1}\tilde{\ell}(\tilde{\boldsymbol{e}}_{z,k+i|k},\bar{\boldsymbol{\tau}}_{k+i|k})+\tilde{V}_f(\tilde{\boldsymbol{e}}_{z,k+N_p|k})\\ \text{subject to}\quad & \tilde{z}_{k+i+1|k}=\tilde{\boldsymbol{A}}\tilde{z}_{k+i|k}+\tilde{\boldsymbol{B}}\bar{\boldsymbol{\tau}}_{k+i|k}+\tilde{\boldsymbol{E}}z_{\mathrm{cmd},k+i|k},\\ & \tilde{z}_{k|k}=\tilde{z}_k,\\ & \tilde{z}_{k+i|k}\in\tilde{\mathbb{Z}},\quad \bar{\boldsymbol{\tau}}_{k+i|k}\in\bar{\mathbb{T}},\\ & \tilde{z}_{k+N_p|k}\in\tilde{\mathbb{Z}}_f\subseteq\tilde{\mathbb{Z}}\end{aligned} \quad (58)$$

where $\tilde{\boldsymbol{e}}_{z,k+i|k}=\tilde{z}_{k+i|k}-\tilde{z}_{\mathrm{cmd},k+i|k}$ is the augmented tracking error, and $\tilde{z}_{\mathrm{cmd},k}=\left[z_{\mathrm{cmd},k}^\top,\ \boldsymbol{0}^\top\right]^\top$ is the augmented lifted-state command. The augmented quadratic stage and terminal costs are defined as $\tilde{\ell}(\tilde{\boldsymbol{e}}_z,\bar{\boldsymbol{\tau}})=\tilde{\boldsymbol{e}}_z^\top\tilde{\boldsymbol{Q}}\tilde{\boldsymbol{e}}_z+\bar{\boldsymbol{\tau}}^\top\boldsymbol{R}\bar{\boldsymbol{\tau}}$ and $\tilde{V}_f(\tilde{\boldsymbol{e}}_z)=\tilde{\boldsymbol{e}}_z^\top\tilde{\boldsymbol{P}}\tilde{\boldsymbol{e}}_z$, respectively, with the augmented weighting matrix is defined as $\tilde{\boldsymbol{Q}}=\mathrm{blockdiag}(\boldsymbol{Q},\boldsymbol{Q}_I)\succeq 0$, where $\boldsymbol{Q}_I\in\mathbb{R}^{N_z\times N_z}$ is the integral gain matrix. The terminal penalty $\tilde{\boldsymbol{P}}\succ 0$ is obtained by solving the DARE for the augmented pair $(\tilde{\boldsymbol{A}},\tilde{\boldsymbol{B}})$ with weights $\tilde{\boldsymbol{Q}}$ and $\boldsymbol{R}$.

The augmented state constraint is defined as

$$\tilde{\mathbb{Z}}=\mathbb{Z}\times\boldsymbol{\Xi}_z,$$

where $\boldsymbol{\Xi}_z=\{\boldsymbol{\xi}_z\in\mathbb{R}^{N_z}\mid \boldsymbol{\xi}_{z,\min}\le\boldsymbol{\xi}_z\le\boldsymbol{\xi}_{z,\max}\}$ explicitly bounds the integral error state. Furthermore, to guarantee recursive feasibility and local stability, the augmented terminal set $\tilde{\mathbb{Z}}_f\subseteq\tilde{\mathbb{Z}}$ is constructed as the MPI set under the unconstrained augmented LQR law $\bar{\boldsymbol{\tau}}=-\tilde{\boldsymbol{K}}\tilde{\boldsymbol{e}}_z$, with the optimal gain $\tilde{\boldsymbol{K}}=(\boldsymbol{R}+\tilde{\boldsymbol{B}}^\top\tilde{\boldsymbol{P}}\tilde{\boldsymbol{B}})^{-1}\tilde{\boldsymbol{B}}^\top\tilde{\boldsymbol{P}}\tilde{\boldsymbol{A}}$.

Remark 11. The existence of a unique, positive-definite stabilizing solution $\tilde{\boldsymbol{P}}$ to the augmented DARE strictly requires $(\tilde{\boldsymbol{A}},\tilde{\boldsymbol{B}})$ to be stabilizable. □

Assumption 5. The offset-free Koopman-based MPC is feasible, and $\bar{\boldsymbol{\tau}}=-\tilde{\boldsymbol{K}}\tilde{\boldsymbol{e}}_z\in\bar{\mathbb{T}}$ to ensure recursive feasibility and local stability. □

Finally, following the receding-horizon principle, the optimal generalized control torque $\bar{\boldsymbol{\tau}}_k$ is extracted and destandardized using the same procedure as in the nominal Koopman-based MPC.

D. STABILITY ANALYSIS

Theorem 1 (Practical Stability of the Cascade System)**.** Under Assumptions 1–5, the closed-loop Xplorer-mini AUV system, consisting of the outer-loop PIwFF kinematic controller and either the nominal or offset-free inner-loop Koopman-based MPC, is uniformly ultimately bounded

(UUB). Consequently, the tracking errors converge to a bounded neighborhood of the origin.

Proof. See Appendix B. □

VI. SIMULATION AND EXPERIMENTAL RESULTS

A. EXPERIMENTAL SETUP

The experimental platform is the custom-built Xplorer-mini AUV (Fig. 4). Navigation system runs on an Intel NUC, which fuses a Teledyne Doppler Velocity Log (DVL) for body-fixed linear velocity, a VectorNav inertial measurement unit (IMU) for attitude and angular rate, and a Blue Robotics Bar-series depth sensor through an unscented Kalman filter (UKF) [39] to estimate the six-degree-of-freedom pose $\boldsymbol{\eta}$ and body-frame velocity $\boldsymbol{\nu}$. Blue Robotics T200 thrusters are driven via dshot from a Teensy board, which receives commands from an NVIDIA Jetson co-processor that also handles camera and image-processing tasks. The vehicle is powered by two 14.8 V, 7500 mAh Li-Po battery packs. All software is implemented in the Robot Operating System (ROS), with commands issued from a topside ground station via cable during pool trials. Data collection and closed-loop validation were performed at Chulabhorn Walailak Swimming Pool (20 m $\times$ 18 m $\times$ 5 m), Kasetsart University.

B. SYSTEM IDENTIFICATION

The identification maneuver follows the time-varying reference pose $\boldsymbol{\eta}_{\text{ref}}(t)$ in Eq. (18), which is tracked by the commanded torque $\boldsymbol{\tau}_{\text{cmd}}$ in Eq. (20). The command combines the cascaded PID output with the perturbation input $\boldsymbol{\tau}_{\text{prt}}$ defined in Eq. (19). Five endpoint configurations are considered, with $\boldsymbol{\eta}_0 = [0, 0, 0.2, 0, 0, 0]^{\mathrm{T}}$ and $\boldsymbol{\eta}_f = [x_f, y_f, z_f, 0, 0, 0]^{\mathrm{T}}$, where $x_f = y_f \in \{6, 7, 8, 9, 10\}$ m and $z_f = 3$ m. For each endpoint configuration, the vehicle is commanded to perform three forward runs ($\boldsymbol{\eta}_0 \to \boldsymbol{\eta}_f$) and three return runs ($\boldsymbol{\eta}_f \to \boldsymbol{\eta}_0$). Each run consists of a maneuvering phase of duration t_f, during which a sinusoidal weaving motion with amplitude A and n_{zig} cycles is superimposed on the straight-line transit to excite the sway–yaw coupling, followed by a hovering phase at the destination pose to provide a steady-state segment. The terminal depth z_f is selected to maintain sufficient clearance from the sloped pool floor, thereby preserving reliable DVL measurements. In total, the experiment yields $5\times(3+3) = 30$ data-collection runs, which are partitioned into 25 training runs and 5 test runs. The corresponding identification parameters are summarized in Table 1.

Prior to identification, the recorded torque $\boldsymbol{\tau}$ and velocity $\boldsymbol{\nu}$ are conditioned to remove sensor-side artefacts that would otherwise corrupt the regression. A Hampel filter is first applied to $\boldsymbol{\tau}$ to reject isolated spikes caused by transient host–firmware polling mismatches that yield out-of-range RPM-feedback samples. A short moving-average window is then applied to both $\boldsymbol{\nu}$ and $\boldsymbol{\tau}$ to attenuate sensor noise that would otherwise inflate the variance of the DMDc and eDMDc regressions. Identical filter parameters are used on the training and test sets to avoid preprocessing asymmetry.

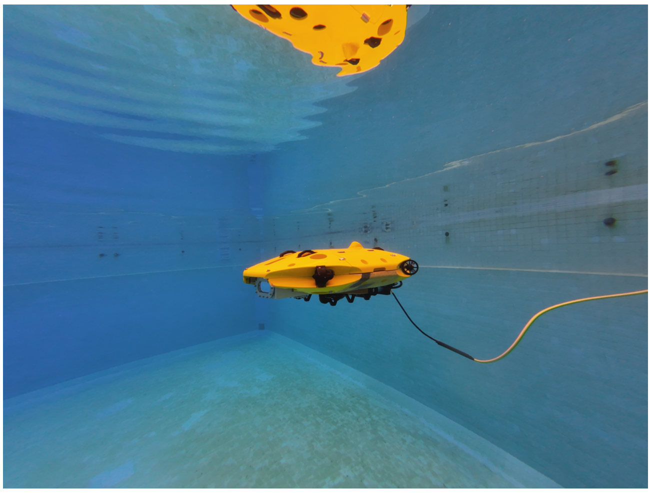

FIGURE 4. The Xplorer-mini AUV platform during a pool trial at Chulabhorn Walailak Swimming Pool, Kasetsart University.

TABLE 1. System Identification Parameters.

Parameter	Value
Initial pose $\boldsymbol{\eta}_0$	$[0,\ 0,\ 0.2,\ 0,\ 0,\ 0]^{\top}$
Terminal pose $\boldsymbol{\eta}_f$	$[x_f,\ y_f,\ z_f,\ 0,\ 0,\ 0]^{\top}$
$x_f = y_f$	$\in \{6, 7, 8, 9, 10\}$ m, five configurations
z_f	3 m
Number of cycles n_{zig}	2
Maneuvering duration t_f	60 s
Hovering duration	40 s
Weaving amplitude A	2 m
Sampling period T_s	0.1 s
Perturbation std. σ	5.0
Trajectory direction	forward ($\boldsymbol{\eta}_0 \to \boldsymbol{\eta}_f$); return ($\boldsymbol{\eta}_f \to \boldsymbol{\eta}_0$)
Total trials	$5 \times (3$ forward $+ 3$ return$) = 30$ runs
Train / test split	25 runs / 5 runs

The identified Koopman-based linear models via DMDc and eDMDc are benchmarked against the first-principles nonlinear 6-DOF AUV model in (6), evaluated with the design-time parameters of the Xplorer-mini AUV. Applying the same protocols to all three models quantifies the predictive gain of the data-driven Koopman approximations relative to the underlying physics-based model under identical conditions.

The predictive accuracy of the identified Koopman models is evaluated on $M = 5$ independent test trajectories under two protocols: one-step prediction and multi-step prediction. Each trajectory contains $N^{(m)} = 1000$ samples, corresponding to a nominal 100-s trial sampled at $T_s = 0.1$ s, with $m \in \{1, \ldots, M\}$ indexing the test trajectory.

a: One-step prediction

At each step k, the model is re-initialized from the ground-truth lifted state $z_k^{(m)}$, so that the prediction error does not accumulate over time. This protocol evaluates the one-step transition accuracy of the identified Koopman model. The resulting single-step forecast is

$$\hat{z}_{k+1}^{(m)} = \boldsymbol{A} z_k^{(m)} + \boldsymbol{B}\,\bar{\boldsymbol{\tau}}_k^{(m)}, \quad k = 1, \ldots, N^{(m)} - 1. \tag{59}$$

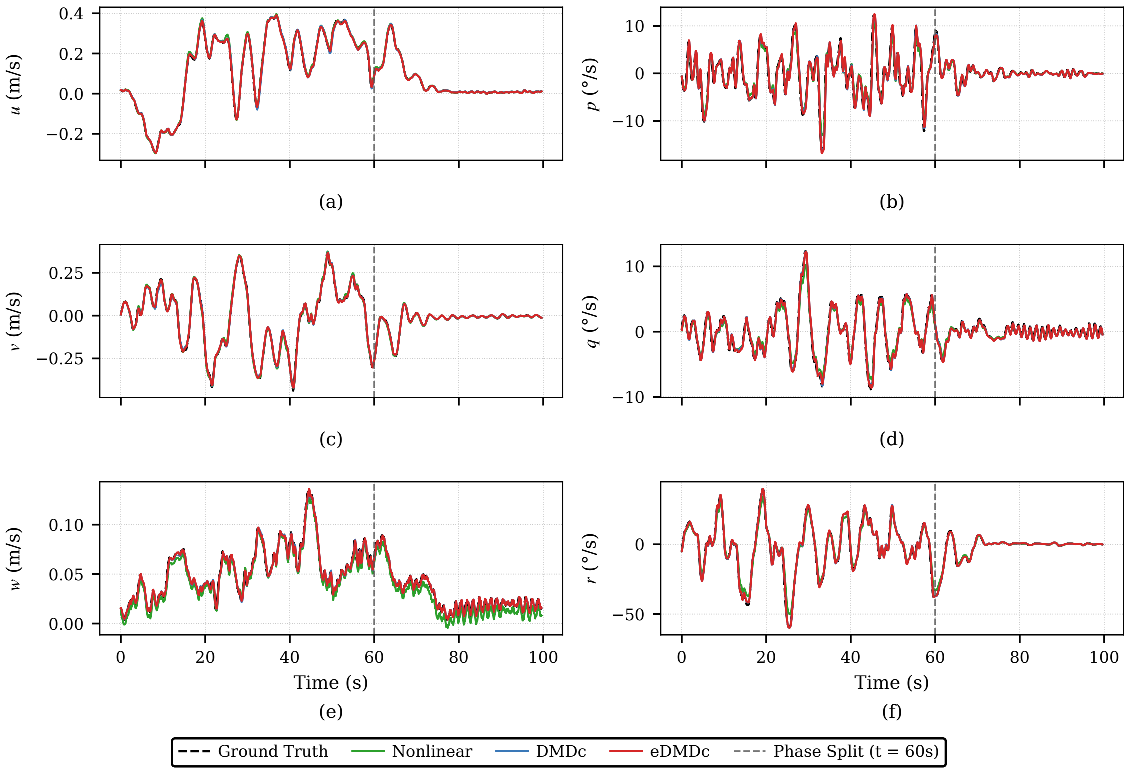

FIGURE 5. **One-step prediction of the Koopman-based models (DMDc and eDMDc) against ground truth for all six velocity states: (a) surge *u*, (b) roll rate *p*, (c) sway *v*, (d) pitch rate *q*, (e) heave *w*, and (f) yaw rate *r*.**

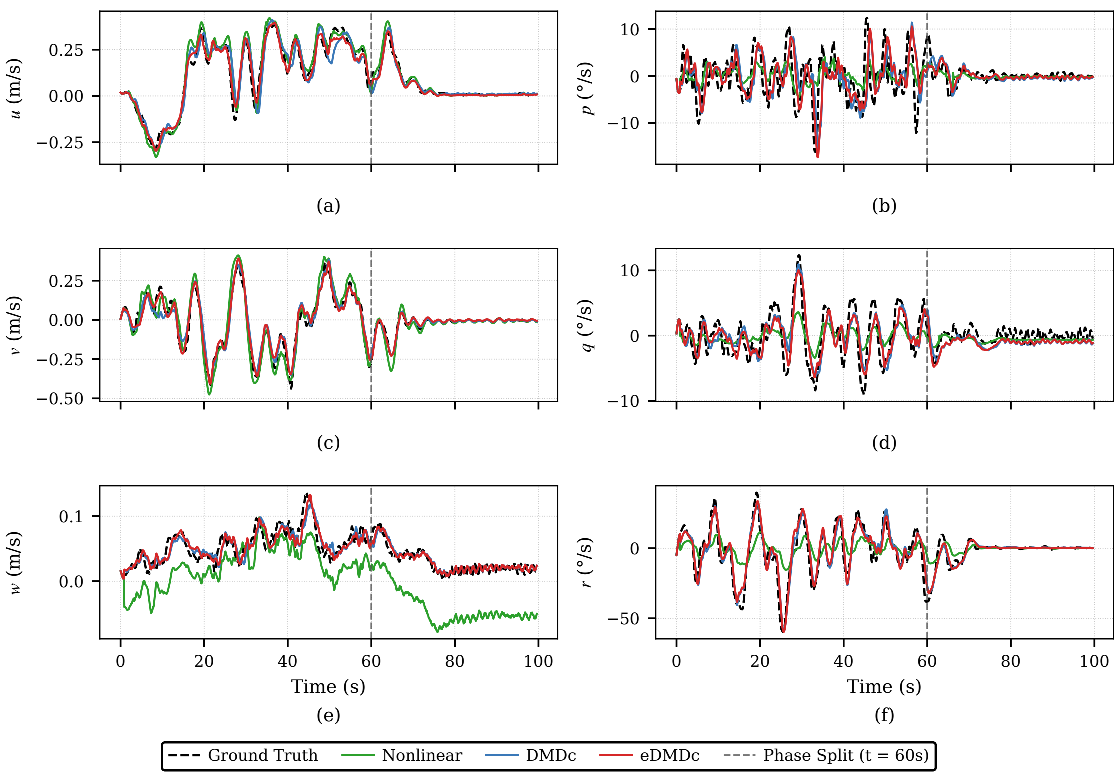

FIGURE 6. **Multi-step prediction ($N_p = 10$) of the Koopman-based models (DMDc and eDMDc) against ground truth for all six velocity states: (a) surge *u*, (b) roll rate *p*, (c) sway *v*, (d) pitch rate *q*, (e) heave *w*, and (f) yaw rate *r*.**

TABLE 2. **Ensemble-averaged RMSE ($\overline{\text{RMSE}}$) of nonlinear, DMDc and eDMDc under one-step and multi-step (N_p = 10) prediction on M = 5 test trajectories, separated by trial phase. Bold values denote the best model per state, horizon, and phase.**

State	Hovering phase						Maneuvering phase					
	One-step prediction			Multi-step prediction			One-step prediction			Multi-step prediction		
	nonlinear	DMDc	eDMDc	nonlinear	DMDc	eDMDc	nonlinear	DMDc	eDMDc	nonlinear	DMDc	eDMDc
u [m/s]	0.0041	0.0035	**0.0027**	0.0259	0.0258	**0.0171**	0.0078	0.0084	**0.0050**	0.0509	0.0640	**0.0271**
v [m/s]	0.0053	0.0045	**0.0041**	0.0335	0.0251	**0.0183**	0.0112	0.0095	**0.0081**	0.0629	0.0537	**0.0369**
w [m/s]	0.0690	**0.0027**	**0.0027**	0.1527	**0.0116**	0.0117	0.0100	0.0033	**0.0032**	0.0888	0.0149	**0.0133**
p [°/s]	0.5371	0.4722	**0.4669**	**2.2452**	2.7252	2.3932	1.0988	0.9180	**0.8861**	4.5829	5.2065	**4.4994**
q [°/s]	0.4869	0.3334	**0.3303**	1.7633	1.6608	**1.5192**	0.6863	0.4397	**0.4129**	2.7734	2.5039	**2.1551**
r [°/s]	1.0862	0.4926	**0.4900**	6.0415	**2.9491**	2.9888	3.5588	1.4701	**1.4407**	17.4561	6.9686	**6.5567**

b: Multi-step prediction

At each prediction index k, the model is re-initialized from the ground-truth lifted state N_p steps earlier, $z^{(m)}_{k-N_p}$, and rolled out over N_p steps using the measured input sequence. The resulting N_p-step-ahead forecast is

$$\hat{z}^{(m)}_k = A^{N_p} z^{(m)}_{k-N_p} + \sum_{s=0}^{N_p-1} A^{N_p-1-s} B\, \bar{\tau}^{(m)}_{k-N_p+s}, \qquad k = N_p+1, \dots, N^{(m)}. \tag{60}$$

The prediction horizon is kept short to limit the error accumulation over the open-loop rollout, since DMDc and eDMDc typically lose accuracy more rapidly over long horizons than deep-learning Koopman variants [37]. Specifically, $N_p = 10$ is used. The same horizon is later used in the Koopman-based MPC formulation

For both protocols, the predicted standardized velocity at each predicted index k is recovered using the linear output map $\hat{\bar{\nu}}^{(m)}_k = C\hat{z}^{(m)}_k$ and is then transformed back to physical units as

$$\hat{\nu}^{(m)}_k = \Sigma_\nu \hat{\bar{\nu}}^{(m)}_k + \mu_\nu, \tag{61}$$

where $\mu_\nu \in \mathbb{R}^6$ and $\Sigma_\nu \in \mathbb{R}^{6\times6}$ are the training-set mean vector and diagonal standard-deviation matrix of the body-fixed velocity, respectively.

To quantify predictive performance, the per-trajectory RMSE for the i-th velocity component is

$$\text{RMSE}^{(m)}_i = \sqrt{\frac{1}{|\mathcal{K}|} \sum_{k\in\mathcal{K}} \left(\nu^{(m)}_{i,k} - \hat{\nu}^{(m)}_{i,k}\right)^2}, \tag{62}$$

where $\mathcal{K} = \{2, \dots, N^{(m)}\}$ for one-step prediction and $\mathcal{K} = \{N_p + 1, \dots, N^{(m)}\}$ for multi-step prediction. The overall accuracy is reported as the ensemble average over all M test trajectories,

$$\overline{\text{RMSE}}_i = \frac{1}{M} \sum_{m=1}^{M} \text{RMSE}^{(m)}_i. \tag{63}$$

During the hovering phase ($t \gtrsim 60$ s), the translational and angular velocities are near zero. For u, v, p, and q, all three models give similar one-step $\overline{\text{RMSE}}$ in Table 2, and their traces in Fig. 5 almost lie on top of one another. The yaw rate r slightly favors the Koopman models, with $\overline{\text{RMSE}}$ roughly half of the nonlinear baseline, but the three models stay in the same order of magnitude. Heave is the one channel that breaks this pattern: the nonlinear baseline drifts to a constant offset in Fig. 5(e), and its heave $\overline{\text{RMSE}}$ is more than an order of magnitude above either Koopman model. Why only heave, and only here? When the vehicle is nearly motionless, the velocity-dependent terms (damping, added mass, Coriolis) are too small to hide a constant mismatch, so the offset is consistent with a static bias. A plausible explanation is water ingress through small cracks in the Xplorer-mini's frame, which shifts $(W - B)$ in $g(\eta)$ (Appendix A) by an unknown amount, while the nonlinear baseline still uses the nominal W from Table 10.

During the maneuvering phase ($0 \leq t \lesssim 60$ s), the vehicle is in motion, and Table 2 shows eDMDc at the lowest one-step $\overline{\text{RMSE}}$ on every channel, DMDc close behind, and the nonlinear baseline trailing on most channels. The largest gap is on yaw rate r: the nonlinear one-step $\overline{\text{RMSE}}$ jumps from 1.09 °/s in hovering to 3.56 °/s here, consistent with the increased contribution of the velocity-dependent terms $C(\nu)\nu$ and $D(\nu)\nu$ in (6). The nonlinear model evaluates these with the nominal hydrodynamic coefficients in Appendix A—inertia and centre-of-gravity from CAD, added mass and damping from manufacturer and CFD-derived values—rather than the effective coefficients of the as-operated Xplorer-mini, producing the large yaw-rate error in Fig. 5(f). The Koopman models, fitted directly to the trial data, partially compensate for these mismatches: the one-step $\overline{\text{RMSE}}$ on r drops to 1.47 °/s and 1.44 °/s for eDMDc. The heave bias from hovering persists in w but is partly masked by the larger dynamic terms.

The prediction results lead to several key observations about the identified Koopman models. First, both DMDc and eDMDc effectively absorb the parameter mismatch of the nominal model, which uses design-time coefficients rather than as-operated ones. In contrast, they yield only an input–output description and do not recover the underlying physical parameters. Second, eDMDc provides a modest open-loop improvement over DMDc because its observables include nonlinear functions of the state, thereby enhancing its ability to capture nonlinear behavior. Third, a remaining residual appears already in one-step prediction and becomes more visible in multi-step prediction due to error accumulation, manifesting as a near-constant offset that a nominal MPC without integral augmentation cannot reject. This motivates the offset-free MPC formulation in Section VI-C.

C. CONTROLLER PERFORMANCE

Closed-loop performance is evaluated on a figure-8 reference path (a Lemniscate of Gerono in the horizontal plane), chosen to test generalization beyond the training trajectories. The vehicle is initialized at $\boldsymbol{\eta}_0 = [0, 0, 0.2, 0, 0, 0]^\top$, and the desired roll and pitch are regulated to $\phi_{\mathrm{ref}} = \theta_{\mathrm{ref}} = 0$ throughout the maneuver $t \in [0, t_f]$. The remaining four reference channels are

$$\begin{aligned} x_{\mathrm{ref}}(t) &= 3.0 - 3.0\cos\vartheta(t), \\ y_{\mathrm{ref}}(t) &= 1.5\sin\big(2\vartheta(t)\big), \\ z_{\mathrm{ref}}(t) &= -3.0, \\ \psi_{\mathrm{ref}}(t) &= \operatorname{atan2}\big(3.0\cos(2\vartheta(t)),\ 3.0\sin\vartheta(t)\big), \end{aligned} \tag{64}$$

where $\vartheta(t)$ denotes a time-varying phase angle that maps the finite interval $t \in [0, t_f]$ onto the angular domain $[0, 2\pi N]$, where N is the number of figure-8 cycles executed within the trial. It is parameterized as $\vartheta(t) = 2\pi N\, s(t)$ using the dimensionless time $s(t) = t/t_f \in [0, 1]$, so that the reference states evolve continuously and at a uniform rate across the horizon [40]. Each trial consists of a 150-s maneuvering phase ($N = 1$ cycle, $t_f = 150$ s) followed by a 40-s hovering phase at the terminal pose to evaluate steady-state regulation.

Before the physical pool experiment, three simulation scenarios were conducted to select and validate the controller configuration. The first scenario compares the DMDc- and eDMDc-identified linear Koopman models as internal prediction models for the nominal Koopman-based MPC using the non-preview reference command. The second scenario evaluates the effect of command preview by comparing the non-preview and preview commands using the selected prediction model. The third scenario compares two PIwFF–Koopman-based MPC controllers against the PID–PID baseline on the figure-8 tracking task. The selected configuration is then deployed in the pool experiment for hardware validation.

In all simulation scenarios, the plant is the full nonlinear 6-DOF AUV model in Eq. (6), with the Xplorer-mini parameters listed in Appendix A. Assuming that the plant–model mismatch is dominated by Koopman approximation residuals, this setup isolates their effect on closed-loop tracking from pool-specific disturbances and reveals the controller's intrinsic tracking performance.

The simulation gains are as follows. The outer-loop PIwFF controller uses $\boldsymbol{K}_{P,\boldsymbol{\eta}} = \mathrm{diag}(0.5,\ 0.5,\ 1.0,\ 1.1,\ 1.31,\ 1.0)$ and $\boldsymbol{K}_{I,\boldsymbol{\eta}} = \mathrm{diag}(0.021,\ 0.036,\ 0.005,\ 0.36,\ 0.43,\ 0.05)$ on $(x, y, z, \phi, \theta, \psi)$, with the integral state bounded by $\boldsymbol{\xi}_{\boldsymbol{\eta},\max} = -\boldsymbol{\xi}_{\boldsymbol{\eta},\min} = \mathbf{1}$ (in metres for translational DOFs, radians for rotational DOFs). This bound clamps the integral term at one unit of accumulated error per DOF, preventing windup under actuator saturation.

For the inner loop, both nominal and offset-free MPC use $\boldsymbol{R} = \mathrm{diag}(1, 1, 1, 1, 1, 1)$. The lifted-state weight matrices for DMDc and eDMDc penalize only the physical velocity states (u, v, w, p, q, r) and not the auxiliary observables, following standard Koopman-MPC practice [36]; accordingly, $\boldsymbol{Q}_{\mathrm{DMDc}} = \mathrm{diag}(100, 250, 50, 50, 50, 100)$ and $\boldsymbol{Q}_{\mathrm{eDMDc}} = \mathrm{blkdiag}(\boldsymbol{Q}_{\mathrm{DMDc}}, \mathbf{0}_{21})$. For the offset-free MPC, the integral penalty is $\boldsymbol{Q}_I = \mathrm{diag}(1, 1, 20, 50, 10, 10)$, giving $\tilde{\boldsymbol{Q}} = \mathrm{blkdiag}(\boldsymbol{Q}, \boldsymbol{Q}_I)$. The inner-loop integral state is bounded by $\boldsymbol{\xi}_{z,\max} = -\boldsymbol{\xi}_{z,\min} = \mathbf{1}$, consistent with the standardized velocity envelope used during system identification. The generalized control limits are $\boldsymbol{\tau}_{\max} = (136.39, 78.75, 138.08, 31.50, 46.07, 12.85)$. These limits are computed from the maximum thrust of the eight Blue Robotics T200 thrusters and the Xplorer-mini thruster allocation matrix.

1) DMDc vs. eDMDc (Simulation)

This case compares the DMDc- and eDMDc-identified linear prediction models within the nominal Koopman-based MPC using the non-preview command, so that any closed-loop difference reflects the prediction model alone. The goal is to evaluate both closed-loop tracking performance and computational suitability for use as the inner-loop prediction model.

In terms of computational efficiency Table 3), DMDc requires a lower mean QP solve time than eDMDc (0.248 ms vs. 0.453 ms), giving a 1.83× speed-up; similar ratios are observed at the median (1.85×) and 95th percentile (1.76×). These statistics are computed over the full 190 s trial at $T_s = 0.1$ s. The speed-up comes from the smaller lifted-state dimension of DMDc, which reduces the QP size at each MPC step. Although both controllers meet the real-time requirement, DMDc provides greater computational headroom for onboard deployment.

As reported in Table 4, the two prediction models provide essentially equivalent velocity-tracking performance. DMDc gives slightly lower RMSE values on five of the six axes (u, v, w, q, and r), whereas eDMDc performs better only on p, where both controllers achieve sub-1°/s tracking error. Since neither model provides a clear tracking advantage, DMDc is selected because of its faster solve time and is used as the prediction model in the subsequent preview-ablation and trajectory-tracking studies.

TABLE 3. Computational Time Statistics of the nominal MPC With Non-Preview Command: DMDc vs. eDMDc as Prediction Model (Simulation).

Metric	DMDc	eDMDc
Mean (ms)	**0.248**	0.453
Std. dev. (ms)	**0.022**	0.031
Median (ms)	**0.242**	0.448
95th percentile (ms)	**0.289**	0.508

TABLE 4. Velocity Tracking RMSE of the nominal MPC With Non-Preview Command: DMDc vs. eDMDc as Prediction Model (Simulation).

State	DMDc	eDMDc
u (m/s)	**0.0081**	0.0084
v (m/s)	**0.0112**	0.0141
w (m/s)	**0.0104**	0.0110
p (°/s)	0.3015	**0.1869**
q (°/s)	**0.2966**	0.3259
r (°/s)	**6.5931**	6.7400

2) Non-Preview vs Preview Commands (Simulation)

On the figure-8 path, only x, y, and ψ are time-varying commands, whereas z, ϕ, and θ are regulated to constants and therefore do not benefit from the preview command. Among the time-varying DOFs, x and y undergo large horizontal-plane excursions with curvature changes near the loop crossings, whereas the heading ψ evolves more smoothly along the path tangent. Therefore, the preview command is expected to provide larger benefits for x and y than for ψ. This case compares the non-preview and preview commands using DMDc as the prediction model.

For position tracking, the worst-case excursion, measured by the maximum absolute error (MaxAE), is the most safety-critical metric because it bounds the largest deviation from the desired path and thus the risk of collision with pool walls or obstacles. As reported in Table 5, the preview command reduces the MaxAE in x and y by approximately 25% and 49%, respectively, while the smoother yaw reference yields a more modest 6% reduction in ψ. The RMSE follows the same trend, with reductions of approximately 18% in x, 39% in y, and 12% in ψ, confirming that preview improves both peak and average tracking on the time-varying DOFs. Based on these safety-oriented improvements, the preview-augmented Koopman-based MPC is selected for the subsequent pool experiments.

TABLE 5. Position Tracking RMSE and MaxAE on Time-Varying DOFs: nominal Koopman-Based MPC without Preview vs. with Preview (Simulation, DMDc).

State	RMSE		MaxAE	
	w/o Preview	w/ Preview	w/o Preview	w/ Preview
x (m)	0.0131	**0.0108**	0.0347	**0.0261**
y (m)	0.0191	**0.0116**	0.0550	**0.0282**
ψ (°)	5.0913	**4.4564**	10.5364	**9.8762**

3) Trajectory Tracking Performance (Simulation)

This scenario compares three controllers on the figure-8 path using the nonlinear 6-DOF simulated plant: PID–PID as the model-free baseline, PIwFF–nominal MPC with preview, and PIwFF–offset-free MPC with preview. Both MPC variants use DMDc as the prediction model.

Across the horizontal-plane channels—x, y, and ψ, which together follow the time-varying figure-8 reference—both PIwFF–MPC variants reduce the tracking RMSE relative to the PID–PID baseline, and the offset-free variant improves further on the nominal one in every case (Table 6). The largest improvement appears on yaw: the RMSE drops from $6.26°$ under PID–PID to $3.44°$ under nominal MPC, and then to $0.41°$ under offset-free MPC.

The depth channel, which is held at a constant reference, shows a different trend. The offset-free MPC again achieves the lowest RMSE, whereas the nominal MPC performs worse than the PID–PID baseline because the Koopman approximation residual appears as a small but persistent steady-state bias, which only the offset-free integrator removes. The MaxAE values in Table 7 preserve the same ordering. The offset-free MPC attains the smallest peak deviation on every channel except roll, where PID–PID leads by about $0.05°$—a margin of no practical significance, since all three controllers keep the roll deviation below $0.2°$.

TABLE 6. Position Tracking RMSE: PID–PID vs. PIwFF–nominal MPC with Preview vs. PIwFF–offset-free MPC with Preview (Simulation, DMDc).

State	PID–PID	PIwFF–nominal	PIwFF–offset-free
x (m)	0.0403	0.0088	**0.0049**
y (m)	0.0442	0.0095	**0.0052**
z (m)	0.0032	0.0113	**0.0002**
ϕ (°)	0.0113	0.0920	**0.0057**
θ (°)	0.0117	0.1148	**0.0038**
ψ (°)	6.2582	3.4380	**0.4085**

TABLE 7. Position Tracking MaxAE: PID–PID vs. PIwFF–nominal MPC with Preview vs. PIwFF–offset-free MPC with Preview (Simulation, DMDc).

State	PID–PID	PIwFF–nominal	PIwFF–offset-free
x (m)	0.0896	0.0261	**0.0099**
y (m)	0.0905	0.0282	**0.0168**
z (m)	0.0328	0.0151	**0.0026**
ϕ (°)	**0.0313**	0.1667	0.0866
θ (°)	0.1071	0.1588	**0.0489**
ψ (°)	16.2401	9.8762	**1.6862**

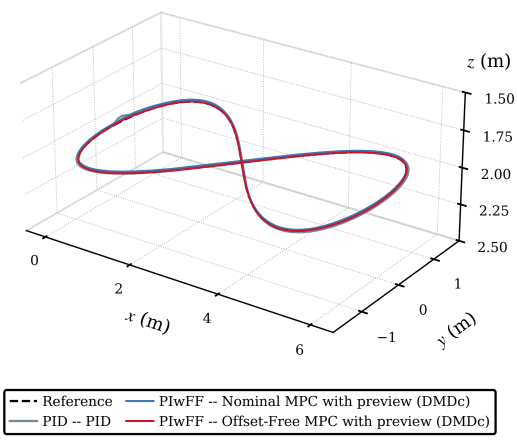

FIGURE 7. Figure-8 3D path (simulation).

The trajectory-level views corroborate the tabulated metrics. In three dimensions, the offset-free MPC traces the figure-8 reference most closely through both lobes (Fig. 7). Fig. 8 resolves the response state by state, and the depth panel (e) provides the clearest separation between the two MPC variants: the offset-free MPC produces a visibly smaller overshoot at the maneuvering-to-hovering transition and then converges to the reference, whereas the nominal MPC leaves a small but visible steady-state offset. Fig. 9 plots the commanded control torques; under both MPC variants, the torques remain strictly inside the actuator bounds for the entire trial, and no constraint ever activates, so the constrained quadratic program stays feasible from start to finish.

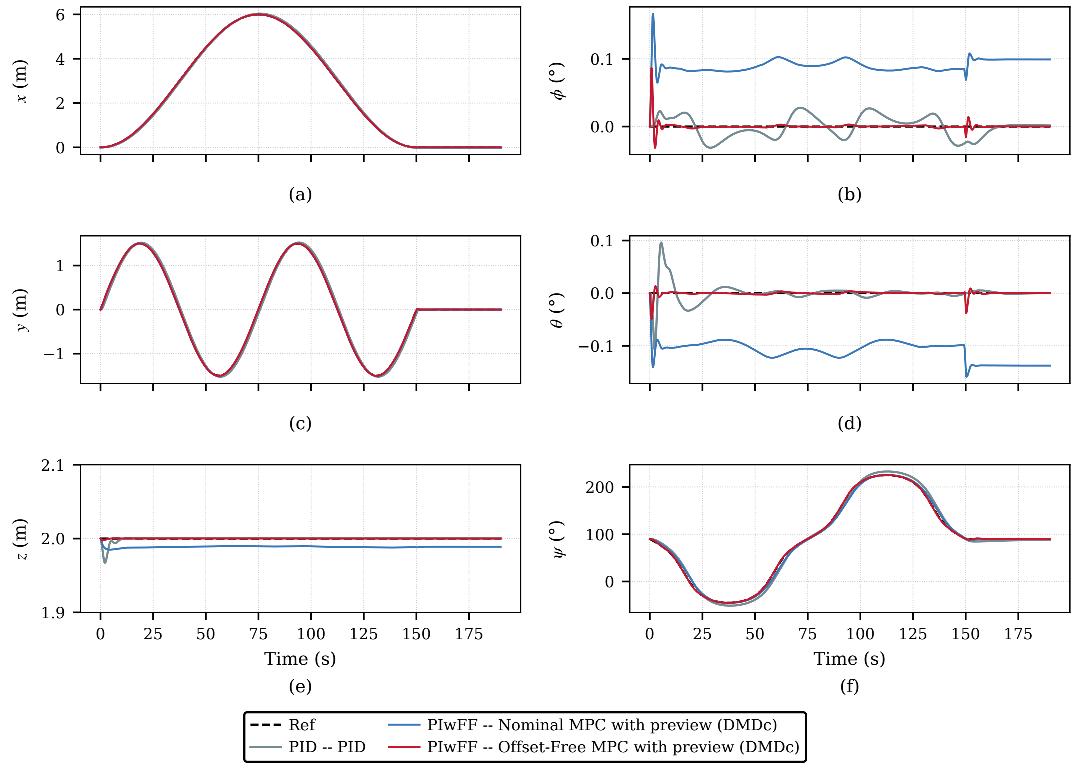

FIGURE 8. Tracking position control response (simulation). Closed-loop position tracking of the Koopman-based MPC controllers (nominal MPC and offset-free MPC with DMDc) against the reference trajectory for all six pose states: (a) surge x, (b) roll ϕ, (c) sway y, (d) pitch θ, (e) heave z, and (f) yaw ψ.

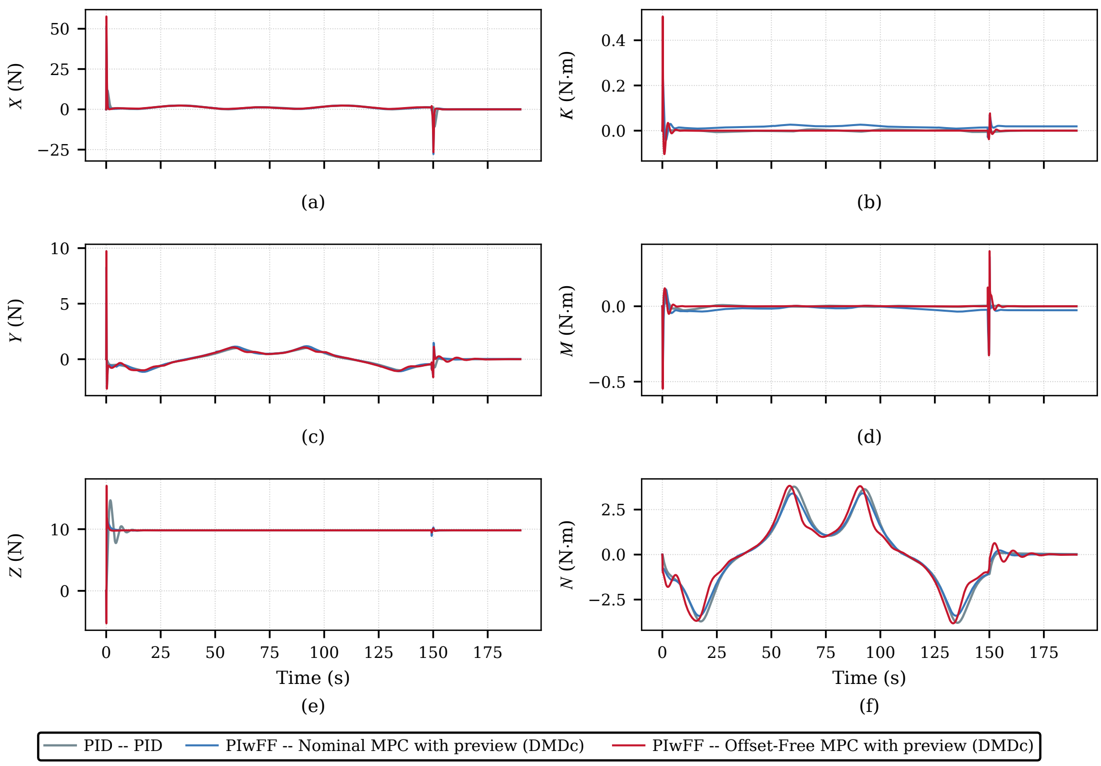

FIGURE 9. Control input trajectories (simulation). Control effort τ of the nominal MPC and offset-free MPC (with preview, DMDc) along the figure-8 path, demonstrating constraint compliance and feasibility of the constrained QP throughout the trial.

4) Trajectory Tracking Performance (Pool Experiment)

Using the same three-controller comparison and figure-8 path, the controllers are now deployed on the physical Xplorer-mini AUV to assess whether the simulation trends carry over to real hardware. The cascaded controller is implemented as a single ROS node, with the outer PIwFF loop and the inner Koopman-based MPC executed synchronously at $f_c = 10$ Hz ($T_s = 0.1$ s).

The controller gains are re-tuned from the simulation values to account for the state estimator and unmodeled water-column disturbances in the physical pipeline of Fig. 2, both of which are absent from the simulated plant. The outer-loop PIwFF controller uses $\boldsymbol{K}_{P,\boldsymbol{\eta}} = \mathrm{diag}(0.5, 0.5, 0.5, 1.1, 1.31, 1.0)$ and $\boldsymbol{K}_{I,\boldsymbol{\eta}} = \mathrm{diag}(0.021, 0.036, 0.01, 0.36, 0.43, 0.05)$. For the inner Koopman-based MPC, the output weight is set to $\boldsymbol{Q}_{\mathrm{DMDc}} = \mathrm{diag}(5.5, 5.5, 1.0, 1.5, 1.5, 5.0)$ and the integral weight to $\boldsymbol{Q}_I = \mathrm{diag}(0.0012, 0.001, 0.0075, 0.005, 0.005, 0.001)$. The input weight $\boldsymbol{R}$, the integral bounds $\boldsymbol{\xi}_{\boldsymbol{\eta}}$ and $\boldsymbol{\xi}_z$, and the control limits $\boldsymbol{\tau}_{\max}$ are kept unchanged from the simulation.

On the horizontal-plane channels, both PIwFF–MPC variants reduce the position-tracking RMSE relative to the PID–PID baseline by roughly one order of magnitude on the translational channels and by a factor of three to four on yaw (Figs. 10 and 11, Table 8). The offset-free MPC gives the lowest RMSE on y and ψ, while the nominal MPC is slightly lower on x by only a few millimetres, a difference with little practical significance over the multi-lap figure-8 trajectory. The MaxAE values in Table 9 show the same overall ranking. This trend is consistent with the simulation study, where the preview-augmented Koopman MPC also provided the best horizontal-plane tracking performance.

In the vertical-plane channels, the two MPC variants differ most clearly in depth: the nominal MPC leaves a near-constant bias, consistent with the water-ingress disturbance of Section IV, whereas the offset-free integrator largely removes it. For roll and pitch, PID–PID gives the lowest RMSE by a small margin because the MPC attitude weights were detuned to avoid estimator noise exciting the integrator; nevertheless, all three controllers maintain sub-degree attitude RMSE. The main benefit of offset-free MPC therefore appears in depth and horizontal tracking, where bias rejection and preview action are most important. As in the simulation (Fig. 9), the commanded torques $\boldsymbol{\tau}$ remain inside the actuator bounds throughout the trial (Fig. 12), confirming that the constrained QP stays feasible under real disturbances.

TABLE 8. Position Tracking RMSE: PID–PID vs. PIwFF–nominal MPC with Preview vs. PIwFF–offset-free MPC with Preview (Pool Experiment, DMDc).

State	PID–PID	PIwFF–nominal	PIwFF–offset-free
x (m)	0.1265	**0.0106**	0.0141
y (m)	0.1483	0.0190	**0.0159**
z (m)	0.0323	0.0327	**0.0057**
ϕ (°)	**0.2747**	0.3264	0.2853
θ (°)	**0.2453**	0.3898	0.5127
ψ (°)	2.6208	0.9111	**0.6941**

TABLE 9. Position Tracking MaxAE: PID–PID vs. PIwFF–nominal MPC with Preview vs. PIwFF–offset-free MPC with Preview (Pool Experiment, DMDc).

State	PID–PID	PIwFF–nominal	PIwFF–offset-free
x (m)	0.2365	0.0371	**0.0353**
y (m)	0.2448	0.0961	**0.0922**
z (m)	0.0828	0.0656	**0.0184**
ϕ (°)	1.0529	0.9866	**0.8624**
θ (°)	**0.8875**	1.1668	1.3740
ψ (°)	6.6799	2.4845	**2.0450**

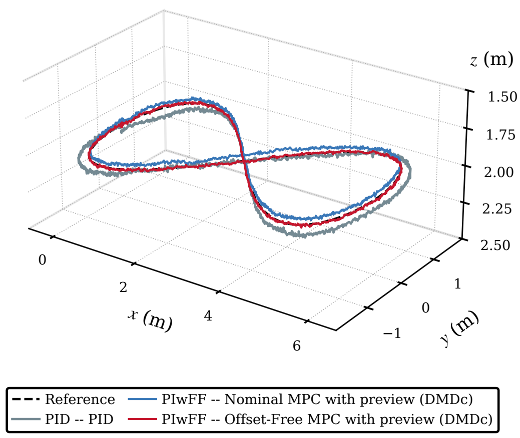

FIGURE 10. Figure-8 3D path (pool).

VII. CONCLUSION

This paper developed and validated a hierarchical PIwFF–Koopman MPC for 6-DOF AUV trajectory tracking on the Xplorer-mini. The proposed architecture uses an outer kinematic PIwFF loop to generate body-fixed velocity references for an inner offset-free Koopman-based linear MPC. The identified Koopman model retains an approximately constant residual on the velocity output, which appears as steady-state error under the nominal MPC even in simulation. In the pool, persistent biases such as buoyancy-related mismatch and sensor offset add to this residual and amplify the same mechanism. The offset-free formulation rejects these effects through an output-integral augmentation, explaining why its advantage over the nominal MPC increases from simulation to pool experiments. The results also show that a richer lifted basis does not necessarily improve closed-loop performance. Although eDMDc provides better open-loop multi-step prediction accuracy, DMDc achieves more reliable closed-loop tracking with a smaller lifted state and faster QP solve time. This result suggests that over-parameterization of the lifted basis can reduce computational efficiency and may not translate into better real-time control performance.

Future work will extend the disturbance model to time-varying perturbations, such as ocean currents, and validate the controller in open-water sea trials.

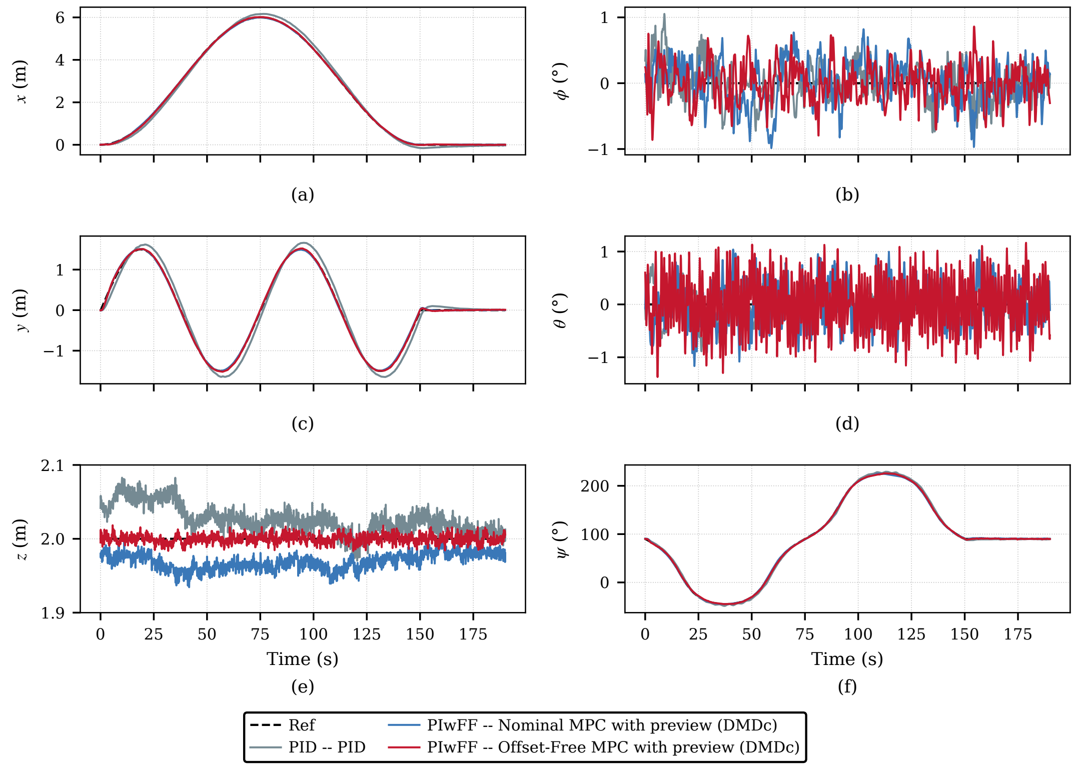

FIGURE 11. Tracking position control response (pool). Closed-loop position tracking of the Koopman-based MPC controllers (nominal MPC and offset-free MPC with DMDc) against the reference trajectory for all six pose states: (a) surge *x*, (b) roll ϕ, (c) sway *y*, (d) pitch θ, (e) heave *z*, and (f) yaw ψ.

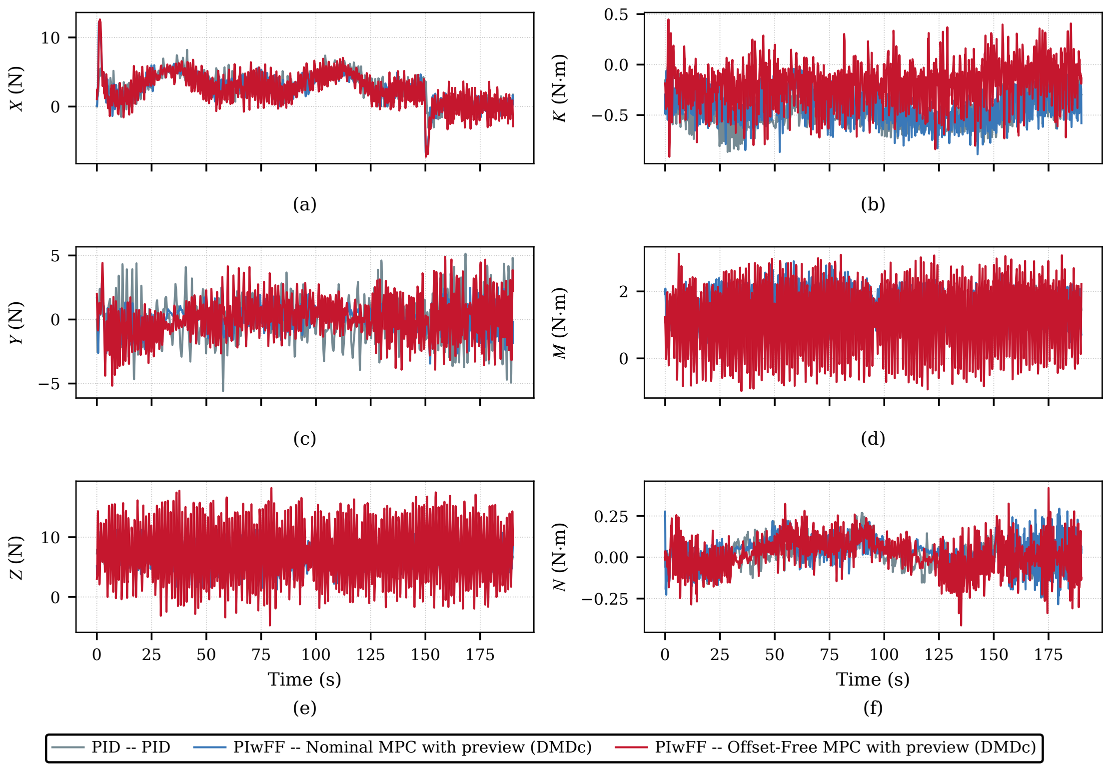

FIGURE 12. Control input trajectories (pool). Control effort τ of the nominal MPC and offset-free MPC (with preview, DMDc) along the figure-8 path, demonstrating constraint compliance and feasibility of the constrained QP under real disturbances.

APPENDIX A
SIMULATION PLANT PARAMETER

This appendix provides the explicit expressions of the inertia matrix $\boldsymbol{M}$, the Coriolis and centripetal matrix $\boldsymbol{C}(\boldsymbol{\nu})$, the damping matrix $\boldsymbol{D}(\boldsymbol{\nu})$, and the restoring vector $\boldsymbol{g}(\boldsymbol{\eta})$ appearing in the 6-DOF nominal dynamics of (6), together with the corresponding numerical parameter values for the Xplorer-mini AUV. The formulation and SNAME notation follow [5]. All plant coefficients are reported symbolically below, with numerical values collected in Table 10; the centres of gravity $\boldsymbol{r}_g$ and buoyancy $\boldsymbol{r}_b$ are expressed in the body-fixed frame.

The total inertia matrix is $\boldsymbol{M} = \boldsymbol{M}_{RB} + \boldsymbol{M}_A$, with the rigid-body inertia

$$\boldsymbol{M}_{RB} = \begin{bmatrix} \boldsymbol{M}_{11} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{M}_{22} \end{bmatrix}, \qquad \boldsymbol{M}_{11} = m\boldsymbol{I}_3, \quad \boldsymbol{M}_{22} = \boldsymbol{I}_b,$$

and the body-fixed inertia tensor

$$\boldsymbol{I}_b = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix}.$$

The added-mass matrix is diagonal,

$$\boldsymbol{M}_A = -\mathrm{diag}\big(X_{\dot u},\, Y_{\dot v},\, Z_{\dot w},\, K_{\dot p},\, M_{\dot q},\, N_{\dot r}\big).$$

The Coriolis and centripetal matrix is decomposed as $\boldsymbol{C}(\boldsymbol{\nu}) = \boldsymbol{C}_{RB}(\boldsymbol{\nu}) + \boldsymbol{C}_A(\boldsymbol{\nu})$. Partition the body-fixed velocity as $\boldsymbol{\nu} = [\boldsymbol{\nu}_1^\top, \boldsymbol{\nu}_2^\top]^\top$ with $\boldsymbol{\nu}_1 = [u, v, w]^\top$ and $\boldsymbol{\nu}_2 = [p, q, r]^\top$, and let $\boldsymbol{S}(\cdot)$ denote the skew-symmetric cross-product operator

$$\boldsymbol{S}(\boldsymbol{a}) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}, \qquad \boldsymbol{a} \in \mathbb{R}^3.$$

Following the skew-symmetric parameterisation, the rigid-body Coriolis matrix is

$$\boldsymbol{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \boldsymbol{0}_{3\times3} & -\boldsymbol{S}(\boldsymbol{M}_{11}\boldsymbol{\nu}_1) \\ -\boldsymbol{S}(\boldsymbol{M}_{11}\boldsymbol{\nu}_1) & -\boldsymbol{S}(\boldsymbol{M}_{22}\boldsymbol{\nu}_2) \end{bmatrix},$$

and the added-mass Coriolis matrix is

$$\boldsymbol{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} \boldsymbol{0}_{3\times3} & -\boldsymbol{S}(\boldsymbol{A}_{11}\boldsymbol{\nu}_1) \\ -\boldsymbol{S}(\boldsymbol{A}_{11}\boldsymbol{\nu}_1) & -\boldsymbol{S}(\boldsymbol{A}_{22}\boldsymbol{\nu}_2) \end{bmatrix},$$

where $\boldsymbol{A}_{11} = -\mathrm{diag}(X_{\dot u}, Y_{\dot v}, Z_{\dot w})$ and $\boldsymbol{A}_{22} = -\mathrm{diag}(K_{\dot p}, M_{\dot q}, N_{\dot r})$ are the diagonal blocks of $\boldsymbol{M}_A$.

The hydrodynamic damping matrix is modelled as a sum of linear and quadratic components,

$$\boldsymbol{D}(\boldsymbol{\nu}) = \boldsymbol{D}_L + \boldsymbol{D}_N\,|\boldsymbol{\nu}|,$$

where, in the standard SNAME notation,

$$\boldsymbol{D}_L = -\mathrm{diag}\big(X_u,\, Y_v,\, Z_w,\, K_p,\, M_q,\, N_r\big),$$

$$\boldsymbol{D}_N = -\mathrm{diag}\big(X_{u|u|},\, Y_{v|v|},\, Z_{w|w|},\, K_{p|p|},\, M_{q|q|},\, N_{r|r|}\big),$$

and $|\boldsymbol{\nu}|$ denotes the element-wise absolute value of the body-fixed velocity. Diagonal damping is adopted under the port/starboard symmetry assumption appropriate for slender AUVs.

The hydrostatic restoring vector, using the shorthand $s_{(\cdot)} = \sin(\cdot)$ and $c_{(\cdot)} = \cos(\cdot)$, is

$$\boldsymbol{g}(\boldsymbol{\eta}) = \begin{bmatrix} (W-B)\,s_\theta \\ -(W-B)\,c_\theta s_\phi \\ -(W-B)\,c_\theta c_\phi \\ -(y_g W - y_b B)\,c_\theta c_\phi + (z_g W - z_b B)\,c_\theta s_\phi \\ (z_g W - z_b B)\,s_\theta + (x_g W - x_b B)\,c_\theta c_\phi \\ -(x_g W - x_b B)\,c_\theta s_\phi - (y_g W - y_b B)\,s_\theta \end{bmatrix},$$

where $W = mg$ and $B = \rho g \nabla$ are the weight and buoyancy magnitudes, and $\boldsymbol{r}_g = [x_g, y_g, z_g]^\top$, $\boldsymbol{r}_b = [x_b, y_b, z_b]^\top$ are the centres of gravity and buoyancy in the body-fixed frame. With $\boldsymbol{r}_g = \boldsymbol{0}$ and $\boldsymbol{r}_b = [0, 0, z_b]^\top$, given Table 10, this reduces to

$$\boldsymbol{g}(\boldsymbol{\eta}) = \begin{bmatrix} (W-B)\,s_\theta \\ -(W-B)\,c_\theta s_\phi \\ -(W-B)\,c_\theta c_\phi \\ -z_b B\,c_\theta s_\phi \\ -z_b B\,s_\theta \\ 0 \end{bmatrix}.$$

TABLE 10. Plant parameters of the Xplorer-mini AUV (SNAME notation).

Description	Symbol	Value	Unit
Physical and geometric			
Mass	m	55.0	kg
Displaced volume	∇	0.05471	m^3
Gravitational acc.	g	9.81	$\mathrm{m/s}^2$
Fluid density	ρ	1024	$\mathrm{kg/m}^3$
Weight	$W = mg$	539.55	N
Buoyancy	$B = \rho g \nabla$	549.36	N
Centre of gravity	$\boldsymbol{r}_g$	$[0, 0, 0]^\top$	m
Centre of buoyancy	$\boldsymbol{r}_b$	$[0, 0, -0.02]^\top$	m
Inertia tensor $\boldsymbol{I}_b$			
	I_{xx}	4.80	$\mathrm{kg\,m}^2$
	I_{yy}	6.30	$\mathrm{kg\,m}^2$
	I_{zz}	2.10	$\mathrm{kg\,m}^2$
	I_{xy}	−0.00598	$\mathrm{kg\,m}^2$
	I_{xz}	0.01147	$\mathrm{kg\,m}^2$
	I_{yz}	0.00081	$\mathrm{kg\,m}^2$
Added-mass coefficients			
	$X_{\dot u}$	−35.10	kg
	$Y_{\dot v}$	−46.80	kg
	$Z_{\dot w}$	−127.80	kg
	$K_{\dot p}$	−14.76	$\mathrm{kg\,m}^2$
	$M_{\dot q}$	−38.22	$\mathrm{kg\,m}^2$
	$N_{\dot r}$	−21.65	$\mathrm{kg\,m}^2$
Linear damping			
	X_u	−0.59	N s/m
	Y_v	−0.05	N s/m
	Z_w	−0.05	N s/m
	K_p	−27.90	N m s/rad
	M_q	−95.25	N m s/rad
	N_r	−24.10	N m s/rad
Quadratic damping			
	$X_{u\|u\|}$	−66.80	$\mathrm{N\,s^2/m^2}$
	$Y_{v\|v\|}$	−87.90	$\mathrm{N\,s^2/m^2}$
	$Z_{w\|w\|}$	−308.30	$\mathrm{N\,s^2/m^2}$
	$K_{p\|p\|}$	−25.50	$\mathrm{N\,m\,s^2/rad^2}$
	$M_{q\|q\|}$	−19.90	$\mathrm{N\,m\,s^2/rad^2}$
	$N_{r\|r\|}$	−20.10	$\mathrm{N\,m\,s^2/rad^2}$

APPENDIX B
STABILITY PROOF

To prove the practical stability of the closed-loop cascade system in Theorem 1, we first establish the ISS properties of the inner-loop Koopman MPC and the outer-loop PIwFF kinematic controller.

Lemma 1. Under Assumptions 1–5, the inner-loop augmented tracking error under the offset-free Koopman MPC is input-to-state stable with respect to the aggregate disturbance $\tilde{\boldsymbol{d}}_k$.

Proof. The disturbed lifted Koopman dynamics are given by

$$\boldsymbol{z}_{k+1} = \boldsymbol{A}\boldsymbol{z}_k + \boldsymbol{B}\bar{\boldsymbol{\tau}}_k + \boldsymbol{w}_k, \qquad \|\boldsymbol{w}_k\| \le w_{\max}. \tag{65}$$

For the offset-free augmented tracking error $\tilde{\boldsymbol{e}}_{z,k}$, the actual error dynamics can be written as

$$\tilde{\boldsymbol{e}}_{z,k+1} = \tilde{\boldsymbol{A}}\tilde{\boldsymbol{e}}_{z,k} + \tilde{\boldsymbol{B}}\bar{\boldsymbol{\tau}}_k + \tilde{\boldsymbol{d}}_k, \tag{66}$$

where the aggregate disturbance is

$$\tilde{\boldsymbol{d}}_k = \begin{bmatrix} \boldsymbol{w}_k + \boldsymbol{A}\boldsymbol{z}_{\mathrm{cmd},k} - \boldsymbol{z}_{\mathrm{cmd},k+1} \\ \boldsymbol{0} \end{bmatrix}. \tag{67}$$

Thus, $\tilde{\boldsymbol{d}}_k$ contains the Koopman approximation residual, environmental disturbances, and the command-variation mismatch. Since the reference trajectory is smooth and the velocity command is saturated, there exists a finite constant $\tilde{d}_{\max} > 0$ such that

$$\|\tilde{\boldsymbol{d}}_k\| \le \tilde{d}_{\max}. \tag{68}$$

Let $\tilde{V}_N(\tilde{\boldsymbol{e}}_{z,k})$ denote the optimal finite-horizon MPC value function. For the nominal augmented system, the standard shifted-sequence MPC argument and the terminal Lyapunov condition imply that

$$\tilde{V}_N\left(\tilde{\boldsymbol{e}}^*_{z,k+1|k}\right) - \tilde{V}_N\left(\tilde{\boldsymbol{e}}_{z,k}\right) \le -\alpha_{\mathrm{i}}\|\tilde{\boldsymbol{e}}_{z,k}\|^2, \tag{69}$$

where $\alpha_{\mathrm{i}} = \lambda_{\min}(\tilde{\boldsymbol{Q}}) > 0$, and $\tilde{\boldsymbol{e}}^*_{z,k+1|k}$ is the nominal one-step-ahead prediction generated by the optimal input at time k.

The actual one-step error satisfies

$$\tilde{\boldsymbol{e}}_{z,k+1} = \tilde{\boldsymbol{e}}^*_{z,k+1|k} + \tilde{\boldsymbol{d}}_k. \tag{70}$$

Let $\tilde{\mathbb{E}}_N$ be a compact subset of the feasible domain of the offset-free MPC value function. Define the disturbance-enlarged compact set as

$$\tilde{\mathbb{E}}_N^{\delta} = \left\{ \tilde{\boldsymbol{e}} + \boldsymbol{\Delta} \;\middle|\; \tilde{\boldsymbol{e}} \in \tilde{\mathbb{E}}_N,\ \|\boldsymbol{\Delta}\| \le \tilde{d}_{\max} \right\}. \tag{71}$$

Under the practical feasibility assumption, $\tilde{\mathbb{E}}_N^{\delta}$ remains contained in the feasible domain of $\tilde{V}_N$. The finite-horizon MPC value function is continuous and piecewise quadratic; hence, it is Lipschitz continuous on $\tilde{\mathbb{E}}_N^{\delta}$. Therefore, there exists $\tilde{L}_V > 0$ such that

$$\left|\tilde{V}_N(\tilde{\boldsymbol{e}}_a) - \tilde{V}_N(\tilde{\boldsymbol{e}}_b)\right| \le \tilde{L}_V\|\tilde{\boldsymbol{e}}_a - \tilde{\boldsymbol{e}}_b\|, \quad \forall \tilde{\boldsymbol{e}}_a, \tilde{\boldsymbol{e}}_b \in \tilde{\mathbb{E}}_N^{\delta}. \tag{72}$$

Applying (72) with $\tilde{\boldsymbol{e}}_a = \tilde{\boldsymbol{e}}^*_{z,k+1|k} + \tilde{\boldsymbol{d}}_k$ and $\tilde{\boldsymbol{e}}_b = \tilde{\boldsymbol{e}}^*_{z,k+1|k}$ gives

$$\tilde{V}_N\left(\tilde{\boldsymbol{e}}_{z,k+1}\right) - \tilde{V}_N\left(\tilde{\boldsymbol{e}}^*_{z,k+1|k}\right) \le \tilde{L}_V\|\tilde{\boldsymbol{d}}_k\|. \tag{73}$$

Combining (69) and (73) yields

$$\tilde{V}_N\left(\tilde{\boldsymbol{e}}_{z,k+1}\right) - \tilde{V}_N\left(\tilde{\boldsymbol{e}}_{z,k}\right) \le -\alpha_{\mathrm{i}}\|\tilde{\boldsymbol{e}}_{z,k}\|^2 + \tilde{L}_V\|\tilde{\boldsymbol{d}}_k\|. \tag{74}$$

Since the value function is positive definite and proper on the compact feasible set, (74) is an ISS-Lyapunov inequality. Therefore, there exist functions $\beta_z \in \mathcal{KL}$ and $\gamma_z \in \mathcal{K}$ such that [41]

$$\|\tilde{\boldsymbol{e}}_{z,k}\| \le \beta_z\left(\|\tilde{\boldsymbol{e}}_{z,0}\|, k\right) + \gamma_z\left(\sup_{0\le i\le k}\|\tilde{\boldsymbol{d}}_i\|\right). \tag{75}$$

Moreover, the physical velocity tracking error is obtained from the lifted tracking error through the bounded output map. In particular,

$$\boldsymbol{e}_{\boldsymbol{\nu},k} = \boldsymbol{\nu}_k - \boldsymbol{\nu}_{\mathrm{cmd},k} = \boldsymbol{\Sigma}_{\nu}\boldsymbol{C}\left(\boldsymbol{z}_k - \boldsymbol{z}_{\mathrm{cmd},k}\right), \tag{76}$$

and hence

$$\|\boldsymbol{e}_{\boldsymbol{\nu},k}\| \le \|\boldsymbol{\Sigma}_{\nu}\boldsymbol{C}[\boldsymbol{I}\ \ \boldsymbol{0}]\|\ \|\tilde{\boldsymbol{e}}_{z,k}\|. \tag{77}$$

Consequently, $\boldsymbol{e}_{\boldsymbol{\nu},k}$ is also input-to-state stable with respect to $\tilde{\boldsymbol{d}}_k$. □

Lemma 2. Under Assumptions 1–5, the outer-loop tracking error under the PIwFF kinematic controller is input-to-state stable with respect to the inner-loop velocity tracking error and the reference discretization residual.

Proof. Define the outer-loop error state as

$$\boldsymbol{e}_{o,k} = \begin{bmatrix} \boldsymbol{e}_{\boldsymbol{\eta},k} \\ \boldsymbol{\xi}_{\boldsymbol{\eta},k} \end{bmatrix}. \tag{78}$$

Using the forward-Euler discretization of the outer-loop kinematics, the outer-loop error dynamics can be written as

$$\boldsymbol{e}_{o,k+1} = \boldsymbol{A}_{\mathrm{o}}\boldsymbol{e}_{o,k} + \boldsymbol{D}_k, \tag{79}$$

where

$$\boldsymbol{A}_{\mathrm{o}} = \begin{bmatrix} \boldsymbol{I} - T_s\boldsymbol{K}_{P,\boldsymbol{\eta}} & -T_s\boldsymbol{K}_{I,\boldsymbol{\eta}} \\ T_s\boldsymbol{I} & \boldsymbol{I} \end{bmatrix} \tag{80}$$

is designed to be Schur by selecting $\boldsymbol{K}_{P,\boldsymbol{\eta}}$ and $\boldsymbol{K}_{I,\boldsymbol{\eta}}$. The perturbation term is

$$\boldsymbol{D}_k = \boldsymbol{B}_{\mathrm{o}}\boldsymbol{J}(\boldsymbol{\eta}_k)\boldsymbol{e}_{\boldsymbol{\nu},k} + \boldsymbol{E}_{\mathrm{o}}\boldsymbol{w}_{\mathrm{ref},k}, \tag{81}$$

with

$$\boldsymbol{B}_{\mathrm{o}} = \begin{bmatrix} -T_s\boldsymbol{I} \\ \boldsymbol{0} \end{bmatrix}, \qquad \boldsymbol{E}_{\mathrm{o}} = \begin{bmatrix} \boldsymbol{I} \\ \boldsymbol{0} \end{bmatrix}. \tag{82}$$

Here, $\boldsymbol{e}_{\boldsymbol{\nu},k} = \boldsymbol{\nu}_k - \boldsymbol{\nu}_{\mathrm{cmd},k}$ is the inner-loop velocity tracking error, and

$$\boldsymbol{w}_{\mathrm{ref},k} = \boldsymbol{\eta}_{\mathrm{ref},k+1} - \boldsymbol{\eta}_{\mathrm{ref},k} - T_s\dot{\boldsymbol{\eta}}_{\mathrm{ref},k} \tag{83}$$

is the reference discretization residual.

Since $\boldsymbol{A}_{\mathrm{o}}$ is Schur, for any $\boldsymbol{Q}_{\mathrm{o}} \succ 0$, there exists a unique $\boldsymbol{P}_{\mathrm{o}} \succ 0$ satisfying

$$\boldsymbol{A}_{\mathrm{o}}^{\top}\boldsymbol{P}_{\mathrm{o}}\boldsymbol{A}_{\mathrm{o}} - \boldsymbol{P}_{\mathrm{o}} = -\boldsymbol{Q}_{\mathrm{o}}. \tag{84}$$

Choose

$$V_{\mathrm{o}}(\boldsymbol{e}_{o,k}) = \boldsymbol{e}_{o,k}^{\top}\boldsymbol{P}_{\mathrm{o}}\boldsymbol{e}_{o,k}. \tag{85}$$

Expanding the Lyapunov difference along (79) gives

$$\begin{aligned} &V_{\mathrm{o}}(\boldsymbol{e}_{o,k+1}) - V_{\mathrm{o}}(\boldsymbol{e}_{o,k}) \\ &\quad = -\boldsymbol{e}_{o,k}^{\top}\boldsymbol{Q}_{\mathrm{o}}\boldsymbol{e}_{o,k} + 2\boldsymbol{e}_{o,k}^{\top}\boldsymbol{A}_{\mathrm{o}}^{\top}\boldsymbol{P}_{\mathrm{o}}\boldsymbol{D}_k + \boldsymbol{D}_k^{\top}\boldsymbol{P}_{\mathrm{o}}\boldsymbol{D}_k. \end{aligned} \tag{86}$$

Using Young's inequality, for any $0 < \varepsilon < \lambda_{\min}(\boldsymbol{Q}_{\mathrm{o}})$,

$$2\boldsymbol{e}_{o,k}^{\top}\boldsymbol{A}_{\mathrm{o}}^{\top}\boldsymbol{P}_{\mathrm{o}}\boldsymbol{D}_k \le \varepsilon\|\boldsymbol{e}_{o,k}\|^2 + \frac{\|\boldsymbol{A}_{\mathrm{o}}^{\top}\boldsymbol{P}_{\mathrm{o}}\|^2}{\varepsilon}\|\boldsymbol{D}_k\|^2.$$

Therefore,

$$V_{\mathrm{o}}(\boldsymbol{e}_{o,k+1}) - V_{\mathrm{o}}(\boldsymbol{e}_{o,k}) \le -c_1\|\boldsymbol{e}_{o,k}\|^2 + c_2\|\boldsymbol{D}_k\|^2, \tag{87}$$

where

$$\begin{aligned} c_1 &= \lambda_{\min}(\boldsymbol{Q}_{\mathrm{o}}) - \varepsilon > 0, \\ c_2 &= \lambda_{\max}(\boldsymbol{P}_{\mathrm{o}}) + \frac{\|\boldsymbol{A}_{\mathrm{o}}^{\top}\boldsymbol{P}_{\mathrm{o}}\|^2}{\varepsilon} > 0. \end{aligned}$$

Since the AUV operates in a bounded region where $\boldsymbol{J}(\boldsymbol{\eta}_k)$ is bounded, there exists $\bar{J} > 0$ such that

$$\|\boldsymbol{J}(\boldsymbol{\eta}_k)\| \le \bar{J}. \tag{88}$$

Hence, from (81),

$$\|\boldsymbol{D}_k\| \le \bar{b}_{\mathrm{o}}\|\boldsymbol{e}_{\boldsymbol{\nu},k}\| + \bar{e}_{\mathrm{o}}\|\boldsymbol{w}_{\mathrm{ref},k}\|, \tag{89}$$

where

$$\bar{b}_{\mathrm{o}} = \|\boldsymbol{B}_{\mathrm{o}}\|\bar{J}, \qquad \bar{e}_{\mathrm{o}} = \|\boldsymbol{E}_{\mathrm{o}}\|. \tag{90}$$

Since $\boldsymbol{P}_{\mathrm{o}} \succ 0$, V_{o} is positive definite and proper. Together with (87) and (89), this establishes an ISS-Lyapunov inequality. Therefore, there exist functions $\beta_{\mathrm{o}} \in \mathcal{KL}$ and $\gamma_{\mathrm{o},\nu}, \gamma_{\mathrm{o},r} \in \mathcal{K}$ such that [41]

$$\begin{aligned} \|\boldsymbol{e}_{o,k}\| \le &\beta_{\mathrm{o}}\left(\|\boldsymbol{e}_{o,0}\|, k\right) + \gamma_{\mathrm{o},\nu}\left(\sup_{0\le i\le k}\|\boldsymbol{e}_{\boldsymbol{\nu},i}\|\right) \\ &+ \gamma_{\mathrm{o},r}\left(\sup_{0\le i\le k}\|\boldsymbol{w}_{\mathrm{ref},i}\|\right). \end{aligned} \tag{91}$$

Thus, the outer-loop tracking error is input-to-state stable with respect to $\boldsymbol{e}_{\boldsymbol{\nu},k}$ and $\boldsymbol{w}_{\mathrm{ref},k}$. □

We now prove the practical stability of the closed-loop cascade system.

Proof. By Lemma 1, the inner-loop velocity tracking error $\boldsymbol{e}_{\boldsymbol{\nu},k}$ is ISS with respect to the aggregate disturbance $\tilde{\boldsymbol{d}}_k$. By Lemma 2, the outer-loop tracking error is ISS with respect to $\boldsymbol{e}_{\boldsymbol{\nu},k}$ and $\boldsymbol{w}_{\mathrm{ref},k}$. Since the proposed architecture is a strict cascade interconnection, the closed-loop cascade system is ISS. Therefore, for the global tracking error

$$\boldsymbol{\chi}_k = \begin{bmatrix} \boldsymbol{e}_{o,k} \\ \boldsymbol{e}_{\boldsymbol{\nu},k} \end{bmatrix}, \tag{92}$$

there exist functions $\beta_c \in \mathcal{KL}$ and $\gamma_c \in \mathcal{K}$ such that

$$\|\boldsymbol{\chi}_k\| \le \beta_c\left(\|\boldsymbol{\chi}_0\|, k\right) + \gamma_c\left(\sup_{0\le i\le k}\|\boldsymbol{\mathcal{W}}_i\|\right), \tag{93}$$

where

$$\boldsymbol{\mathcal{W}}_i = \begin{bmatrix} \tilde{\boldsymbol{d}}_i \\ \boldsymbol{w}_{\mathrm{ref},i} \end{bmatrix}. \tag{94}$$

By Assumption 1 and the boundedness of the smooth reference trajectory, there exists a finite constant $\bar{W} > 0$ such that

$$\sup_{k\ge 0}\|\boldsymbol{\mathcal{W}}_k\| \le \bar{W}. \tag{95}$$

It follows from (93) that

$$\|\boldsymbol{\chi}_k\| \le \beta_c\left(\|\boldsymbol{\chi}_0\|, k\right) + \gamma_c(\bar{W}). \tag{96}$$

Since $\beta_c(\|\boldsymbol{\chi}_0\|, k) \to 0$ as $k \to \infty$,

$$\limsup_{k\to\infty}\|\boldsymbol{\chi}_k\| \le \gamma_c(\bar{W}). \tag{97}$$

Therefore, all closed-loop tracking errors are uniformly ultimately bounded.

Finally, since the physical pose tracking error is a component of the outer-loop error state,

$$\|\boldsymbol{e}_{\boldsymbol{\eta},k}\| \le \|\boldsymbol{e}_{o,k}\| \le \|\boldsymbol{\chi}_k\|, \tag{98}$$

the physical pose tracking error is also uniformly ultimately bounded and converges to a bounded neighborhood of the origin. □

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